SIS SAMPLE DRAFT — FULL TIER

Nassau County Sample

Traffic Impact Study

Prepared September 2, 2026 · 40.8623, -73.6337

SIS sample draft — full tier

Traffic Impact Study

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TRAFFIC IMPACT STUDY FOR DOT PROJECTS

Prepared in Accordance With Chapter 5 of the NYSDOT Highway Design Manual (HDM)

Project name	Nassau County Sample
PIN	PIN xxxx.xx (to be assigned by NYSDOT)
Project description	Shopping Center / Retail (85 1,000 sqft GLA) within New York-Newark-Jersey City MSA.
NYSDOT Region	Region 10 — Long Island
Site coordinates	40.8623°, -73.6337°
Opening year (ETC)	2027

Note:

This report is based on the NYSDOT TIS Shell revised on 9/16/2014.

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PROJECT CONTEXT

NYSDOT region: Region 10 — Long Island
Planning group: NYMTC (New York Metropolitan Transportation Council)
Regulatory regime: SEQRA (ECL Article 8 / 6 NYCRR Part 617).
SEQRA classification: Not yet determined — lead agency to classify as Type I (6 NYCRR Part 617.4 — likely EIS), Type II (Part 617.5 — exempt), or Unlisted (Part 617.6 — Short EAF + significance finding) before locking the TIS.

1.0 SUMMARY OF TRAFFIC IMPACTS

Capacity:

A capacity analysis was performed per NYSDOT Highway Design Manual Chapter 5. The following capacity improvement measures are cost effective, and are recommended to reduce reoccurring delay: Moderate: extend critical-phase green time and consider a protected-only left-turn phase to absorb the new demand without queue spillback; Moderate: extend critical-phase green time and consider a protected-only left-turn phase to absorb the new demand without queue spillback; Moderate: extend critical-phase green time and consider a protected-only left-turn phase to absorb the new demand without queue spillback; Moderate: extend critical-phase green time and consider a protected-only left-turn phase to absorb the new demand without queue spillback; Moderate: extend critical-phase green time and consider a protected-only left-turn phase to absorb the new demand without queue spillback; Minor: signal-timing optimization (shift 3–5s of green to the critical phase) is sufficient. No geometric change required.

Crash:

A minimum three-year accident history review was performed: 155,492 crashes county-wide in NASSAU County over the rolling 3-year window (NY DMV via data.ny.gov). Section 4.0 carries the severity breakdown and HDM Chapter 5 §5.3.4 screening context; the project-specific rate comparison against NYSDOT statewide averages remains a submittal-stage task with Regional Traffic Office data.

GML §239-m / §239-n county planning board referral required:

The site is within 500 ft of GLEN COVE AVE (Urban Minor Arterial), which triggers mandatory referral to the controlling county planning board under General Municipal Law §239-m (site-plan / special-permit / variance / zoning amendment) or §239-n (subdivision review) before the local board may act. The county may make adverse findings under §239-f; the local board can override only by supermajority. Practical TIS consequence: the county will typically request the TIS in full as part of the §239 referral package — pre-application coordination with the county planning agency is recommended.

22	6	6	22.6s
INTERSECTIONS	LOS DROPS	AT LOS E/F	WORST Δ DELAY

2.0 TRAVEL SPEEDS

Per NYSDOT HDM Chapter 5 §5.2, posted speed limits and actual operating speeds (85th-percentile) along the study network are reported below. Posted-speed-limit values in the third column are auto-ingested from the NYSDOT Roadway Data Management (RDM) FeatureServer (https://gis.dot.ny.gov/hostingny/rest/services/Roadways/RDM_Roadway_Current/FeatureServer) at PDF-generation time — the same source data viewable through the NYSDOT Traffic Data Viewer (https://gis.dot.ny.gov/html5viewer/?viewer=tdv). Operating speeds (85th-percentile) are NOT published by NYSDOT and require a project-specific radar / floating-car speed study.

Street	Functional class	Posted speed	85th-pctile operating speed
Glen Cove Avenue (GLEN COVE AVE)	Urban Minor Arterial	30 mph	Speed study required
Glen Cove Avenue	Refer to NYSDOT TDV	—	Speed study required
Glen Cove Avenue	Refer to NYSDOT TDV	—	Speed study required
Herb Hill Road (GLEN COVE AVE)	Urban Minor Arterial	30 mph	Speed study required
Morris Avenue (GLEN COVE AVE)	Urban Minor Arterial	30 mph	Speed study required
Herb Hill Road (CHARLES ST)	Urban Major Collector	30 mph	Speed study required

3.0 CAPACITY ANALYSIS

Capacity analyses in this report use the SimpleImpactStudies screening solver — a per-approach control-delay model built on the openly-published Webster uniform-delay and Akçelik overflow formulations — reported against the conventional US control-delay bands. Synchro / SimTraffic, HCS, Sidra, or VISSIM may be substituted for formal submittal per NYSDOT Regional Traffic Office preference.

Study network functional-class composition (per NYSDOT RDM): Urban Minor Arterial (11); Urban Major Collector (5); Urban Principal Arterial Other (2); Urban Local (2). NYSDOT functional class drives access management expectations under HDM Appendix 5A (entrance standards for state highways) and the default K-factor banding under HDM §5.3.

Level of Service thresholds applied below are the conventional US control-delay and density bands for signalized, unsignalized (2-way stop / all-way stop / roundabout), and freeway facilities, as republished in publicly-available state DOT design manuals. Any component v/c > 1.0 is LOS F regardless of delay or density.

Signalized intersections — conventional US control-delay bands, sec/veh (as republished in state DOT design manuals); any v/c > 1.0 is F.

LOS	A	B	C	D	E	F
sec/veh	≤10	10–20	20–35	35–55	55–80	>80

Unsignalized — 2-way stop / all-way stop / roundabout — conventional US control-delay bands, sec/veh; any v/c > 1.0 is F.

LOS	A	B	C	D	E	F

LOS	A	B	C	D	E	F
sec/veh	≤10	10–15	15–25	25–35	35–50	>50

Freeway basic / weaving / merge-diverge — conventional US density bands, pc/mi/ln; any component vd/c > 1.00 is F.

LOS	A	B	C	D	E	F
pc/mi/ln	≤11	11–18	18–26	26–35	35–45	>45

3.1 Growth Rates

Background traffic growth is applied at 1.5% per year over 1 year. This rate is the per-station median CAGR measured from the NYSDOT Traffic Monitoring AADT FeatureServer for Region 10 — Long Island: 0.07%/yr median across 12,244 count stations (rolling actual-count window approximately 2000-2019; 25th-75th percentile range -1.57% to 1.80%). The measured value supersedes the NYSDOT statewide default fallbacks (per HDM Chapter 5: 1.7%/yr Interstate + ramps from the Data Services Bureau, 2.0%/yr all other roads from the Regional Planning Group). The controlling Regional Planning Group NYMTC (New York Metropolitan Transportation Council) may have separately-adopted growth-rate guidance that overrides the measured median for formal submittal — confirm at scoping.

3.2 Existing AADT and DHV

Existing AADT is sourced from the NYSDOT Traffic Data Viewer. Design Hourly Volume (DHV) is reported per HDM Chapter 5 convention as AADT × K-factor; this analysis applies K = 0.10 as the statewide design default — facility-specific K should be substituted from NYSDOT count-station records for formal submittal. The table below grows the existing volume at 1.5%/yr to Estimated Time of Completion (ETC), ETC + 20 years, and ETC + 30 years per HDM Chapter 5 §5.3.

Intersection	2026 AADT	2026 DHV	2027 (ETC) AADT	2027 (ETC) DHV	2047 AADT	2047 DHV	2057 AADT	2057 DHV
Glen Cove Avenue & Charles Street	17,130	1,713	17,387	1,739	23,418	2,342	27,177	2,718
Glen Cove Avenue & Brewster Street	17,091	1,709	17,347	1,735	23,364	2,336	27,115	2,712
Glen Cove Avenue & Continental Place	3,049	305	3,095	309	4,168	417	4,837	484
Herb Hill Road & Brewster Street	17,130	1,713	17,387	1,739	23,418	2,342	27,177	2,718
Morris Avenue & Glen Cove Avenue	17,130	1,713	17,387	1,739	23,418	2,342	27,177	2,718
Herb Hill Road & Charles Street	1,170	117	1,188	119	1,599	160	1,856	186
Bridge Street & School Street	3,061	306	3,107	311	4,185	418	4,856	486
Pratt Boulevard & Pulaski Street	10,199	1,020	10,352	1,035	13,943	1,394	16,181	1,618
Glen Street & Pulaski Street	3,060	306	3,106	311	4,183	418	4,855	485
Brewster Street & School Street	10,451	1,045	10,608	1,061	14,287	1,429	16,581	1,658
School Street & Highland Road	1,001	100	1,016	102	1,368	137	1,588	159
Shore Road & Glen Cove Avenue	17,131	1,713	17,388	1,739	23,419	2,342	27,179	2,718
Glen Street & Knoll Place	3,060	306	3,106	311	4,183	418	4,855	485
Pratt Boulevard & Town Path	11,750	1,175	11,926	1,193	16,063	1,606	18,642	1,864
School Street & Brewster Street	3,670	367	3,725	373	5,017	502	5,823	582
Landing Road & Roosevelt Street	3,580	358	3,634	363	4,894	489	5,680	568
Glen Street & Town Path	3,060	306	3,106	311	4,183	418	4,855	485
Burns Avenue & Glen Cove Avenue	17,130	1,713	17,387	1,739	23,418	2,342	27,177	2,718
Dosoris Lane & Forest Avenue	1,150	115	1,167	117	1,572	157	1,825	182
Glen Street & Hendrick Avenue	1,649	165	1,674	167	2,254	225	2,616	262

Intersection	2026 AADT	2026 DHV	2027 (ETC) AADT	2027 (ETC) DHV	2047 AADT	2047 DHV	2057 AADT	2057 DHV
Forest Avenue & Elliott Place	14,960	1,496	15,184	1,518	20,451	2,045	23,734	2,373
Cedar Swamp Road & Elm Avenue	18,610	1,861	18,889	1,889	25,441	2,544	29,525	2,953

Source: AADT estimated from engine peak-hour entering-volume sum via K = 0.10. NYSDOT Traffic Data Viewer measured AADT is recommended for submittal. Projection applies the 1.5%/yr background growth rate compounded annually.

3.3 Traffic Control Device Data

Existing traffic control devices at each study-network intersection are summarized below. Signal timing data (cycle length, phase splits, offset) and per-approach detection presence should be confirmed against the controlling NYSDOT Regional Traffic Office or municipal traffic-engineering signal record for formal submittal.

Intersection	Control type	Approaches w/ detection
Glen Cove Avenue & Charles Street	Signal (controlling RTO to confirm)	Field verification required
Glen Cove Avenue & Brewster Street	Signal (controlling RTO to confirm)	Field verification required
Glen Cove Avenue & Continental Place	Signal (controlling RTO to confirm)	Field verification required
Herb Hill Road & Brewster Street	Signal (controlling RTO to confirm)	Field verification required
Morris Avenue & Glen Cove Avenue	Signal (controlling RTO to confirm)	Field verification required
Herb Hill Road & Charles Street	Signal (controlling RTO to confirm)	Field verification required
Bridge Street & School Street	Signal (controlling RTO to confirm)	Field verification required
Pratt Boulevard & Pulaski Street	Signal (controlling RTO to confirm)	Field verification required
Glen Street & Pulaski Street	Signal (controlling RTO to confirm)	Field verification required
Brewster Street & School Street	Signal (controlling RTO to confirm)	Field verification required
School Street & Highland Road	Signal (controlling RTO to confirm)	Field verification required
Shore Road & Glen Cove Avenue	Signal (controlling RTO to confirm)	Field verification required
Glen Street & Knoll Place	Signal (controlling RTO to confirm)	Field verification required
Pratt Boulevard & Town Path	Signal (controlling RTO to confirm)	Field verification required
School Street & Brewster Street	Signal (controlling RTO to confirm)	Field verification required
Landing Road & Roosevelt Street	Signal (controlling RTO to confirm)	Field verification required
Glen Street & Town Path	Signal (controlling RTO to confirm)	Field verification required
Burns Avenue & Glen Cove Avenue	Signal (controlling RTO to confirm)	Field verification required
Dosoris Lane & Forest Avenue	Signal (controlling RTO to confirm)	Field verification required
Glen Street & Hendrick Avenue	Signal (controlling RTO to confirm)	Field verification required
Forest Avenue & Elliott Place	Signal (controlling RTO to confirm)	Field verification required
Cedar Swamp Road & Elm Avenue	Signal (controlling RTO to confirm)	Field verification required

3.4 Capacity Analysis for Existing No-Build Condition

Existing No-Build conditions reflect current-year peak-hour volumes (no growth, no project trips). Existing measured volumes are grown to opening year 2027 at 1.5%/yr to establish the No-Build baseline for §3.5 comparison.

Intersection	Existing LOS	Existing delay (s)	No-Build LOS	No-Build delay (s)	v/c
Glen Cove Avenue & Charles Street	D	45.6	D	48.4	0.97
Glen Cove Avenue & Brewster Street	D	45.2	D	48.0	0.96
Glen Cove Avenue & Continental Place	B	15.2	B	15.2	0.17

Intersection	Existing LOS	Existing delay (s)	No-Build LOS	No-Build delay (s)	v/c
Herb Hill Road & Brewster Street	D	45.6	D	48.4	0.97
Morris Avenue & Glen Cove Avenue	D	45.6	D	48.4	0.97
Herb Hill Road & Charles Street	B	14.2	B	14.2	0.07
Bridge Street & School Street	B	15.2	B	15.2	0.17
Pratt Boulevard & Pulaski Street	C	21.1	C	21.3	0.58
Glen Street & Pulaski Street	B	15.2	B	15.2	0.17
Brewster Street & School Street	C	21.5	C	21.7	0.59
School Street & Highland Road	B	14.1	B	14.1	0.06
Shore Road & Glen Cove Avenue	D	45.6	D	48.4	0.97
Glen Street & Knoll Place	B	15.2	B	15.2	0.17
Pratt Boulevard & Town Path	C	23.3	C	23.6	0.66
School Street & Brewster Street	B	15.6	B	15.6	0.21
Landing Road & Roosevelt Street	B	15.5	B	15.5	0.20
Glen Street & Town Path	B	15.2	B	15.2	0.17
Burns Avenue & Glen Cove Avenue	D	45.6	D	48.4	0.97
Dosoris Lane & Forest Avenue	B	14.2	B	14.2	0.06
Glen Street & Hendrick Avenue	B	14.4	B	14.4	0.09
Forest Avenue & Elliott Place	C	31.4	C	32.4	0.84
Cedar Swamp Road & Elm Avenue	E	65.2	E	69.9	1.05

Trip Distribution by Time of Day

Estimated within-day distribution of the 6,800 gross daily trips and the resulting on-site accumulation. Within-day distribution from NHTS 2022 shopping / errand / meal trip start times, cross-checked against public-data time-of-day patterns for LU 820 / 850. Activity builds to a midday plateau and a stronger late-afternoon/evening peak; on-site occupancy peaks midday through the evening.

Figure — Inbound / Outbound Trips by Start Time

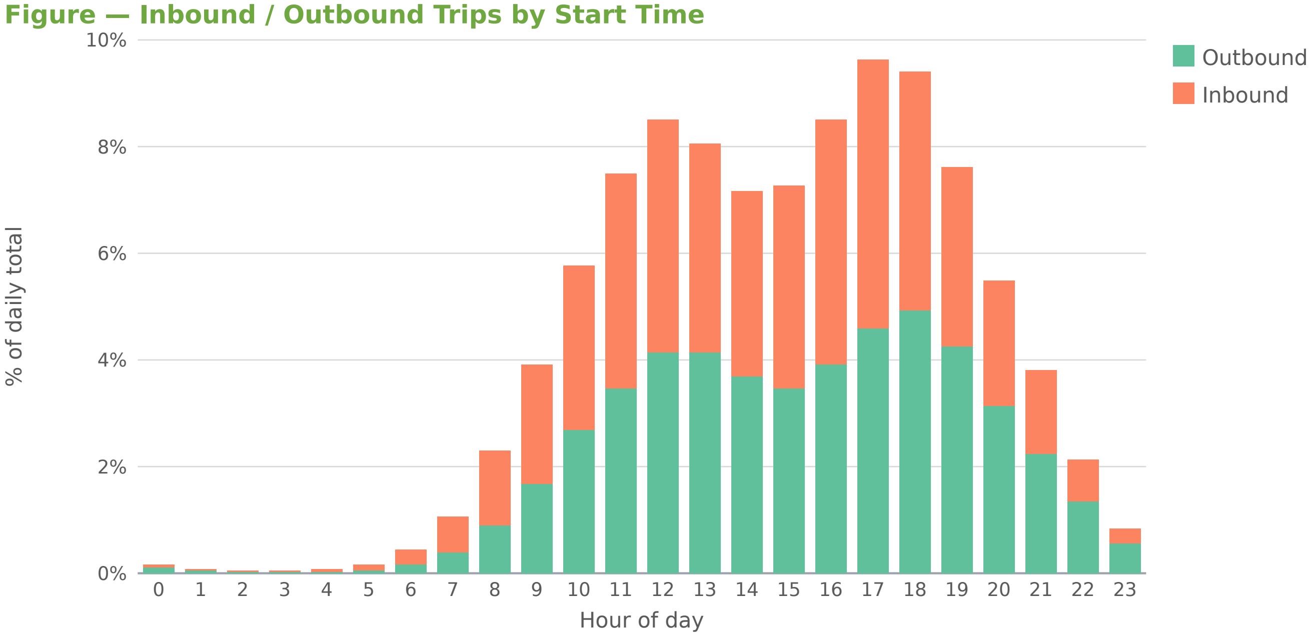
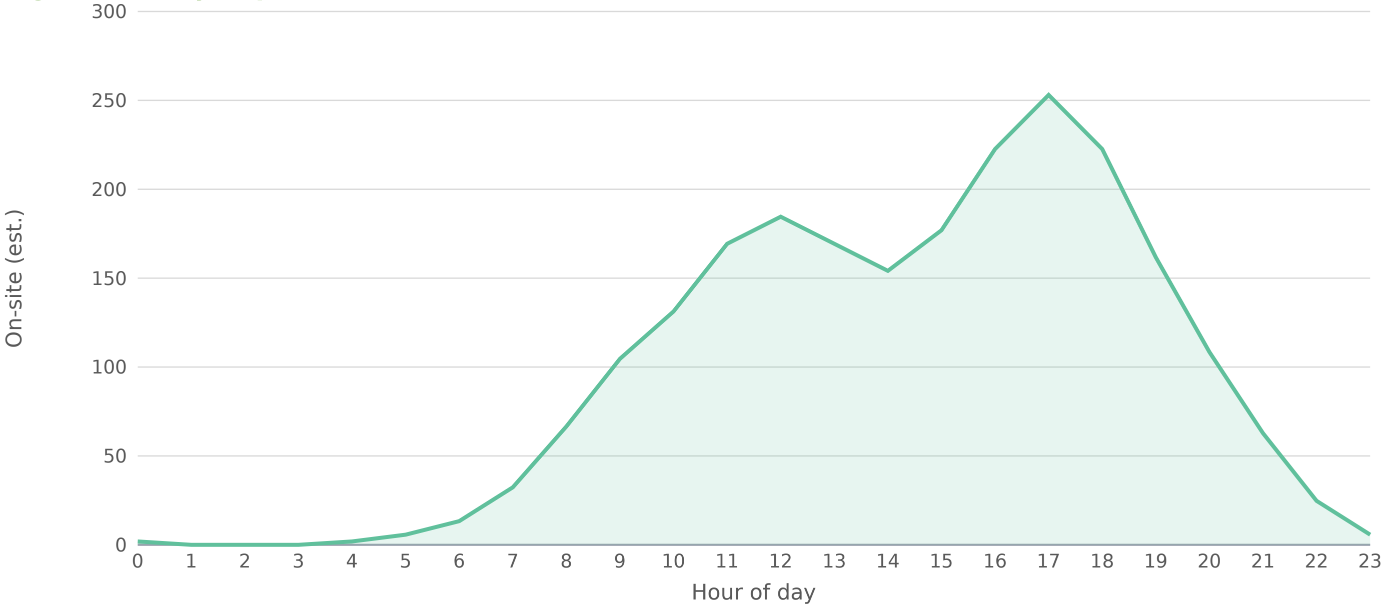


Figure — Daily Trip Accumulation



Peak on-site accumulation ~253 at 17:00, from 6,800 gross daily trips on the retail within-day profile. Screening estimate; not a substitute for a calibrated time-of-day model.

3.5 Capacity Analysis for Proposed Build Condition

Build conditions add the project's external peak-hour trips (680 PM peak, 326 inbound / 354 outbound) to the No-Build baseline at each affected intersection. Per HDM Chapter 5, intersection projects should also analyze each adjacent intersection plus any signal within ½ mile of the project site if expected left-turn volume changes materially affect capacity. The engine's network is determined by a 0.75-mile screening radius; manual scoping should confirm the ½-mile left-turn rule is satisfied.

Intersection	ETC No-Build	ETC Build	ETC+20 NB	ETC+20 Bld	Δ delay (s)
Glen Cove Avenue & Charles Street	D	▲ E	F	F	22.6
Glen Cove Avenue & Brewster Street	D	▲ E	F	F	14.0
Glen Cove Avenue & Continental Place	B	B	B	B	0.3
Herb Hill Road & Brewster Street	D	▲ E	F	F	7.9
Morris Avenue & Glen Cove Avenue	D	▲ E	F	F	7.7
Herb Hill Road & Charles Street	B	B	B	B	0.4
Bridge Street & School Street	B	B	B	B	0.5
Pratt Boulevard & Pulaski Street	C	C	C	C	0.3
Glen Street & Pulaski Street	B	B	B	B	0.3
Brewster Street & School Street	C	C	C	C	0.5
School Street & Highland Road	B	B	B	B	0.2
Shore Road & Glen Cove Avenue	D	▲ E	F	F	7.7
Glen Street & Knoll Place	B	B	B	B	0.1
Pratt Boulevard & Town Path	C	C	D	D	0.4
School Street & Brewster Street	B	B	B	B	0.4
Landing Road & Roosevelt Street	B	B	B	B	0.1

Intersection	ETC No-Build	ETC Build	ETC+20 NB	ETC+20 Bld	Δ delay (s)
Glen Street & Town Path	B	B	B	B	0.2
Burns Avenue & Glen Cove Avenue	D	D	F	F	3.4
Dosoris Lane & Forest Avenue	B	B	B	B	0.3
Glen Street & Hendrick Avenue	B	B	B	B	0.1
Forest Avenue & Elliott Place	C	▲ D	F	F	3.1
Cedar Swamp Road & Elm Avenue	E	E	F	F	2.3

3.6 Capacity Improvement Measures

The following capacity improvements are recommended at affected intersections under Build conditions. Improvement category guidance per HDM Chapter 5: addition of turn lanes; installation or re-timing of traffic signals; signage and striping to prevent unwanted turning movements; conversion to roundabout; and TDM / TSM measures per HDM Chapter 24.

Glen Cove Avenue & Charles Street [MODERATE]
Moderate: extend critical-phase green time and consider a protected-only left-turn phase to absorb the new demand without queue spillback.

Glen Cove Avenue & Brewster Street [MODERATE]
Moderate: extend critical-phase green time and consider a protected-only left-turn phase to absorb the new demand without queue spillback.

Herb Hill Road & Brewster Street [MODERATE]
Moderate: extend critical-phase green time and consider a protected-only left-turn phase to absorb the new demand without queue spillback.

Morris Avenue & Glen Cove Avenue [MODERATE]
Moderate: extend critical-phase green time and consider a protected-only left-turn phase to absorb the new demand without queue spillback.

Shore Road & Glen Cove Avenue [MODERATE]
Moderate: extend critical-phase green time and consider a protected-only left-turn phase to absorb the new demand without queue spillback.

Burns Avenue & Glen Cove Avenue [MINOR]
Minor: signal-timing optimization (shift 3–5s of green to the critical phase) is sufficient. No geometric change required.

Forest Avenue & Elliott Place [MINOR]
Minor: signal-timing optimization (shift 3–5s of green to the critical phase) is sufficient. No geometric change required.

Cedar Swamp Road & Elm Avenue [MODERATE]
Moderate: extend critical-phase green time and consider a protected-only left-turn phase to absorb the new demand without queue spillback.

3.7 Trip Distribution — Gravity Model

The trip distribution uses the gravity model (term = M / (d^1 · d_site); mass basis: intersection through-volume (AADT × K-factor) as the destination-activity attraction proxy). Shares are normalized to 100% of project trips and drive the trip assignment below. Basis: NCHRP-716 gamma-friction gravity model (production-constrained) over study-area attraction zones.

Directional sector	Share of project trips
Northeast (NNE + ENE)	47%
Southeast (ESE + SSE)	15%

Directional sector	Share of project trips
Southwest (SSW + WSW)	36%
Northwest (WNW + NNW)	2%

Study-area zone	Dir.	Distance (mi)	Mass (M)	Gravity term	Trip share
Glen Cove Avenue & Charles Street	WSW	0.01	1,713	0.1	9.68%
Glen Cove Avenue & Brewster Street	NNE	0.04	1,709	0.1	9.66%
Herb Hill Road & Brewster Street	NNE	0.10	1,713	0.1	9.46%
Morris Avenue & Glen Cove Avenue	SSW	0.14	1,713	0.1	9.29%
Shore Road & Glen Cove Avenue	SSW	0.37	1,713	0.1	8.52%
Cedar Swamp Road & Elm Avenue	ESE	0.72	1,861	0.1	8.23%
Burns Avenue & Glen Cove Avenue	SSW	0.50	1,713	0.1	8.14%
Forest Avenue & Elliott Place	NNE	0.67	1,496	0.1	6.73%
Pratt Boulevard & Town Path	ESE	0.42	1,175	0.1	5.75%
Pratt Boulevard & Pulaski Street	ENE	0.22	1,020	0.1	5.36%
Brewster Street & School Street	NNE	0.30	1,045	0.1	5.32%
School Street & Brewster Street	NNE	0.43	367	0.0	1.79%
Landing Road & Roosevelt Street	NNW	0.44	358	0.0	1.74%
Glen Cove Avenue & Continental Place	ENE	0.10	305	0.0	1.69%
Bridge Street & School Street	NNE	0.16	306	0.0	1.65%
Glen Street & Pulaski Street	ENE	0.24	306	0.0	1.60%
Glen Street & Knoll Place	ENE	0.41	306	0.0	1.50%
Glen Street & Town Path	ENE	0.50	306	0.0	1.46%
Glen Street & Hendrick Avenue	ESE	0.58	165	0.0	0.76%
Herb Hill Road & Charles Street	WNW	0.16	117	0.0	0.63%

22 study-area zone(s) (top 20 by trip share shown). Screening-grade distribution; not a calibrated regional model.

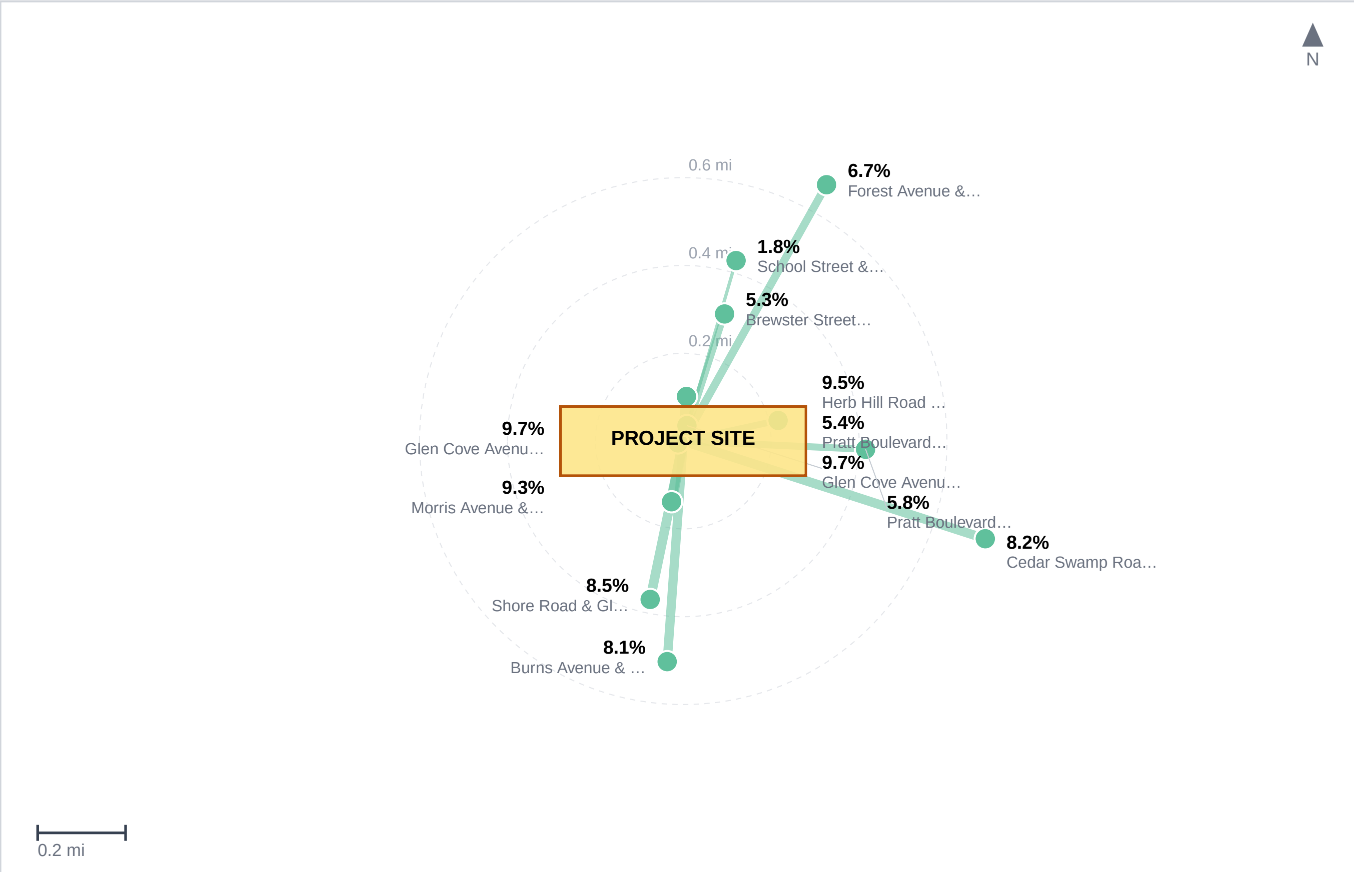
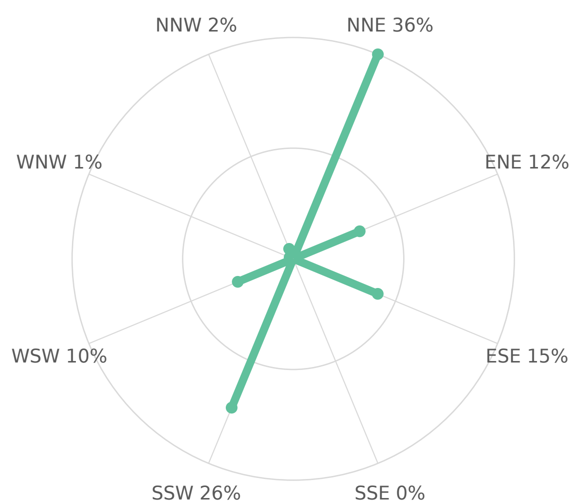


Figure — Project Trip Distribution. Study-area zones plotted to scale at their true bearing and distance from the site; leg weight is proportional to each zone's share of project trips, and the label gives that share. Derived from the Gravity Model distribution — the same shares tabulated above. Screening-grade: zone positions are the analysis locations, not a surveyed base map.

Figure — Directional Distribution of Project Trips



Screening-grade directional distribution of net new project trips by compass octant (spoke length ∝ percent of project trips). Derived from the Gravity Model distribution.

Figure — Project Trip Share by Study-Area Zone

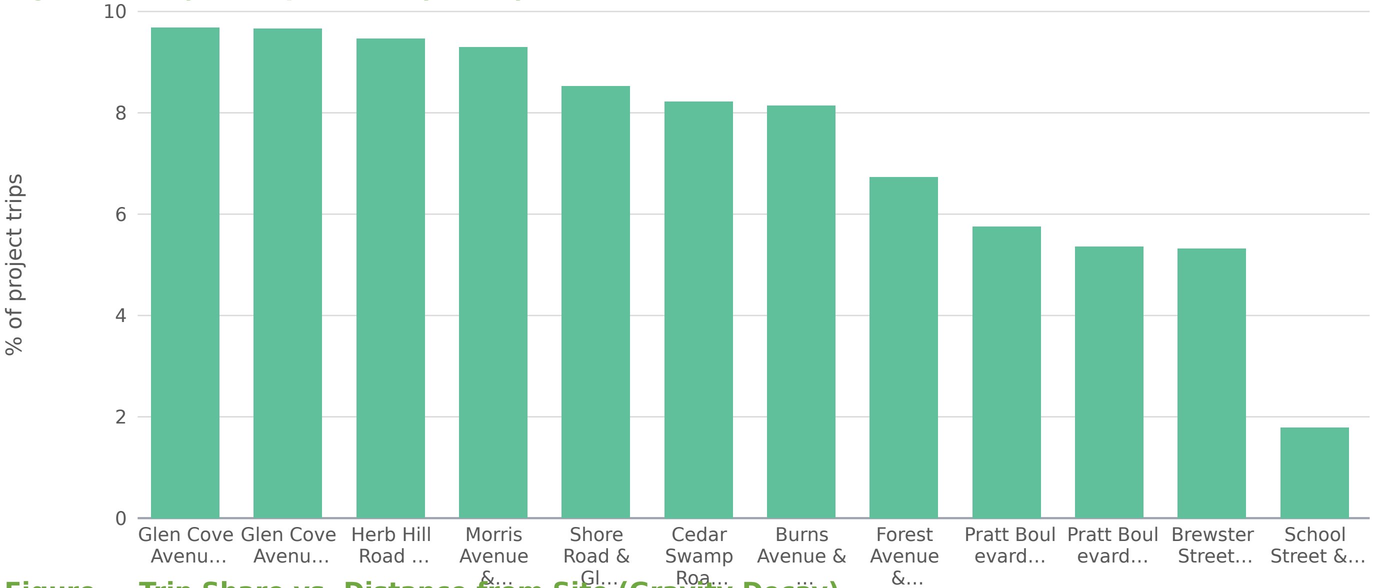
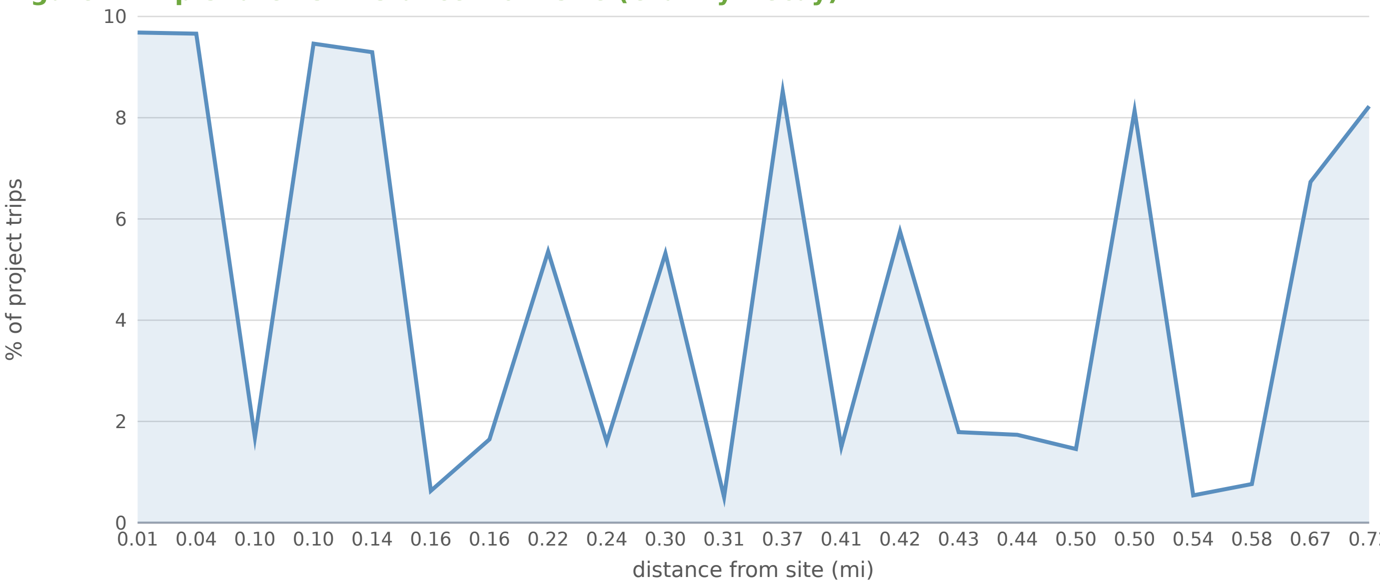


Figure — Trip Share vs. Distance from Site (Gravity Decay)



3.8 Project Trip Assignment

Study intersection	Dir.	Distance (mi)	AM trips	PM trips	Share of project
Glen Cove Avenue & Charles Street	WSW	0.01	83	156	14.0%
Glen Cove Avenue & Brewster Street	NNE	0.04	56	106	9.5%
Glen Cove Avenue & Continental Place	ENE	0.10	23	44	3.9%
Herb Hill Road & Brewster Street	NNE	0.10	33	62	5.6%
Morris Avenue & Glen Cove Avenue	SSW	0.14	32	60	5.4%
Herb Hill Road & Charles Street	WNW	0.16	40	75	6.7%

Study intersection	Dir.	Distance (mi)	AM trips	PM trips	Share of project
Bridge Street & School Street	NNE	0.16	39	75	6.7%
Pratt Boulevard & Pulaski Street	ENE	0.22	9	23	2.1%
Glen Street & Pulaski Street	ENE	0.24	27	51	4.6%
Brewster Street & School Street	NNE	0.30	19	36	3.2%
School Street & Highland Road	NNE	0.31	18	35	3.1%
Shore Road & Glen Cove Avenue	SSW	0.37	32	60	5.4%
Glen Street & Knoll Place	ENE	0.41	11	21	1.9%
Pratt Boulevard & Town Path	ESE	0.42	9	23	2.1%
School Street & Brewster Street	NNE	0.43	33	62	5.6%
Landing Road & Roosevelt Street	NNW	0.44	10	18	1.6%
Glen Street & Town Path	ENE	0.50	18	31	2.8%
Burns Avenue & Glen Cove Avenue	SSW	0.50	15	28	2.5%
Dosoris Lane & Forest Avenue	NNE	0.54	33	62	5.6%
Glen Street & Hendrick Avenue	ESE	0.58	5	13	1.2%
Forest Avenue & Elliott Place	NNE	0.67	33	62	5.6%
Cedar Swamp Road & Elm Avenue	ESE	0.72	5	13	1.2%

Project trips assigned to each study intersection from the distribution shares above: a four-step gravity model sets the share reaching each intersection from its attraction mass and its distance from the site. Directional load multipliers are a Florida (Caltran) refinement and are not applied in this region — each intersection's loading is taken directly from the gravity weights above.

4.0 CRASH ANALYSIS

A minimum three-year accident history review was performed for the period 2023 to 2026 per HDM Chapter 5 §5.3.4. County-level crash counts below are auto-ingested from the New York State Police "Motor Vehicle Crashes – Case Information: Three Year Window" Open Data dataset (https://data.ny.gov/Transportation/Motor-Vehicle-Crashes-Case-Information-Three-Year-/e8ky-4vqe), filtered by county_name resolved from the project site coordinates via the NYSDOT RDM_Roadway_Current FeatureServer. The values cover the entire rolling 3-year reporting window for NASSAU County, statewide-system-wide — they are NOT a per-segment or per-intersection rate.

Severity descriptor (NY DMV)	Count (3-yr rolling)	Annualized
Fatal Accident	280	93.3
Injury Accident	6,376	2125.3
Property Damage Accident	113,811	37937.0
Property Damage & Injury Accident	35,025	11675.0
Total (all severities)	155,492	51830.7

Within NASSAU County over the rolling 3-year window, 155,492 crashes were reported to NY DMV (280 fatal, 41,401 injury-involved, 113,811 property damage only). The Fatal share is 0.060% of all reported crashes (annualized). For the Safety Screening per HDM Chapter 5 §5.3.4, the project-specific rate must be normalized against the controlling facility's exposure (segment AADT × segment length; intersection AADT × entry-flow) for comparison to NYSDOT Office of Modal Safety statewide averages and the 1.5× HAL trigger. County-totals above provide a county-baseline reference only — site-radius (1/4-mile typical) crash records, HAL / PIL / SDL flags, and CMF / CRF mitigation values must be obtained from the Region 10 — Long Island Regional Traffic Office via FOIL or SIMS. Required NYSDOT forms when the §4.0 section is expanded for formal submittal: TE-156a (Collision Diagram), TE-164a (Safety Benefits Evaluation), TE-204a (Accident Rate Summary), TE-213 (HAL / PIL / SDL Screening Output).

APPENDIX A — EXISTING VOLUME REPORT

Per-approach existing peak-hour volumes (vph) for all affected intersections within study limits. Source: SimpleImpactStudies screening solver (Webster/Akçelik control-delay model), calibrated against the controlling DOT 511 / Regional Traffic Office data feed.

Intersection	Approach	Existing (vph)	No-Build (vph)	Build (vph)
Glen Cove Avenue & Charles Street	NB	533	541	588
Glen Cove Avenue & Charles Street	SB	442	449	449
Glen Cove Avenue & Charles Street	EB	453	460	488
Glen Cove Avenue & Charles Street	WB	284	289	370
Glen Cove Avenue & Brewster Street	NB	523	531	586
Glen Cove Avenue & Brewster Street	SB	430	437	467
Glen Cove Avenue & Brewster Street	EB	432	438	438
Glen Cove Avenue & Brewster Street	WB	324	329	350
Glen Cove Avenue & Continental Place	NB	89	90	90
Glen Cove Avenue & Continental Place	SB	78	79	79
Glen Cove Avenue & Continental Place	EB	81	82	105
Glen Cove Avenue & Continental Place	WB	57	58	79
Herb Hill Road & Brewster Street	NB	486	493	526
Herb Hill Road & Brewster Street	SB	462	469	498
Herb Hill Road & Brewster Street	EB	417	423	423
Herb Hill Road & Brewster Street	WB	349	354	354
Morris Avenue & Glen Cove Avenue	NB	529	537	566
Morris Avenue & Glen Cove Avenue	SB	441	448	479
Morris Avenue & Glen Cove Avenue	EB	413	419	419
Morris Avenue & Glen Cove Avenue	WB	329	334	334
Herb Hill Road & Charles Street	NB	39	39	49
Herb Hill Road & Charles Street	SB	28	29	32
Herb Hill Road & Charles Street	EB	27	27	51
Herb Hill Road & Charles Street	WB	23	23	62
Bridge Street & School Street	NB	80	82	120
Bridge Street & School Street	SB	79	80	106
Bridge Street & School Street	EB	81	82	82
Bridge Street & School Street	WB	66	67	77
Pratt Boulevard & Pulaski Street	NB	304	308	308
Pratt Boulevard & Pulaski Street	SB	228	232	232
Pratt Boulevard & Pulaski Street	EB	257	261	284
Pratt Boulevard & Pulaski Street	WB	231	235	235
Glen Street & Pulaski Street	NB	87	89	89
Glen Street & Pulaski Street	SB	79	80	94
Glen Street & Pulaski Street	EB	84	86	112

Intersection	Approach	Existing (vph)	No-Build (vph)	Build (vph)
Glen Street & Pulaski Street	WB	55	56	67
Brewster Street & School Street	NB	336	341	360
Brewster Street & School Street	SB	279	284	298
Brewster Street & School Street	EB	228	232	232
Brewster Street & School Street	WB	202	205	207
School Street & Highland Road	NB	29	29	47
School Street & Highland Road	SB	24	25	38
School Street & Highland Road	EB	24	24	24
School Street & Highland Road	WB	24	24	28
Shore Road & Glen Cove Avenue	NB	555	563	576
Shore Road & Glen Cove Avenue	SB	390	395	427
Shore Road & Glen Cove Avenue	EB	389	395	410
Shore Road & Glen Cove Avenue	WB	380	386	386
Glen Street & Knoll Place	NB	92	93	93
Glen Street & Knoll Place	SB	75	76	81
Glen Street & Knoll Place	EB	73	74	85
Glen Street & Knoll Place	WB	66	67	73
Pratt Boulevard & Town Path	NB	339	344	344
Pratt Boulevard & Town Path	SB	291	295	295
Pratt Boulevard & Town Path	EB	323	328	350
Pratt Boulevard & Town Path	WB	222	226	226
School Street & Brewster Street	NB	114	116	148
School Street & Brewster Street	SB	86	87	117
School Street & Brewster Street	EB	91	92	92
School Street & Brewster Street	WB	77	78	78
Landing Road & Roosevelt Street	NB	100	102	111
Landing Road & Roosevelt Street	SB	100	101	109
Landing Road & Roosevelt Street	EB	87	88	88
Landing Road & Roosevelt Street	WB	71	72	73
Glen Street & Town Path	NB	83	84	94
Glen Street & Town Path	SB	92	94	115
Glen Street & Town Path	EB	71	72	72
Glen Street & Town Path	WB	60	61	61
Burns Avenue & Glen Cove Avenue	NB	519	527	540
Burns Avenue & Glen Cove Avenue	SB	368	373	388
Burns Avenue & Glen Cove Avenue	EB	490	497	497
Burns Avenue & Glen Cove Avenue	WB	336	341	341
Dosoris Lane & Forest Avenue	NB	32	33	33
Dosoris Lane & Forest Avenue	SB	30	31	31

Intersection	Approach	Existing (vph)	No-Build (vph)	Build (vph)
Dosoris Lane & Forest Avenue	EB	30	30	63
Dosoris Lane & Forest Avenue	WB	23	23	53
Glen Street & Hendrick Avenue	NB	51	52	52
Glen Street & Hendrick Avenue	SB	37	37	50
Glen Street & Hendrick Avenue	EB	46	47	47
Glen Street & Hendrick Avenue	WB	31	31	31
Forest Avenue & Elliott Place	NB	434	441	441
Forest Avenue & Elliott Place	SB	368	373	373
Forest Avenue & Elliott Place	EB	362	367	399
Forest Avenue & Elliott Place	WB	333	338	367
Cedar Swamp Road & Elm Avenue	NB	551	559	559
Cedar Swamp Road & Elm Avenue	SB	507	515	528
Cedar Swamp Road & Elm Avenue	EB	442	448	448
Cedar Swamp Road & Elm Avenue	WB	362	367	367

APPENDIX B — EXISTING CONDITION CAPACITY ANALYSIS OUTPUT

Per-intersection screening-solver output (Webster/Akçelik control-delay model) for the Existing No-Build condition (current-year volumes; volumes grown to opening year at the §3.1 background-growth rate for No-Build comparison).

Intersection	Approach	Vol (vph)	v/c	Delay (s)	LOS
Glen Cove Avenue & Charles Street	NB	541	0.67	23.8	C
Glen Cove Avenue & Charles Street	SB	449	0.55	20.9	C
Glen Cove Avenue & Charles Street	EB	460	0.57	21.2	C
Glen Cove Avenue & Charles Street	WB	289	0.36	17.4	B
Glen Cove Avenue & Brewster Street	NB	531	0.66	23.4	C
Glen Cove Avenue & Brewster Street	SB	437	0.54	20.5	C
Glen Cove Avenue & Brewster Street	EB	438	0.54	20.6	C
Glen Cove Avenue & Brewster Street	WB	329	0.41	18.2	B
Glen Cove Avenue & Continental Place	NB	90	0.11	14.6	B
Glen Cove Avenue & Continental Place	SB	79	0.10	14.5	B
Glen Cove Avenue & Continental Place	EB	82	0.10	14.5	B
Glen Cove Avenue & Continental Place	WB	58	0.07	14.2	B
Herb Hill Road & Brewster Street	NB	493	0.61	22.1	C
Herb Hill Road & Brewster Street	SB	469	0.58	21.4	C
Herb Hill Road & Brewster Street	EB	423	0.52	20.2	C
Herb Hill Road & Brewster Street	WB	354	0.44	18.7	B
Morris Avenue & Glen Cove Avenue	NB	537	0.66	23.7	C
Morris Avenue & Glen Cove Avenue	SB	448	0.55	20.8	C
Morris Avenue & Glen Cove Avenue	EB	419	0.52	20.1	C
Morris Avenue & Glen Cove Avenue	WB	334	0.41	18.3	B
Herb Hill Road & Charles Street	NB	39	0.05	14.0	B
Herb Hill Road & Charles Street	SB	29	0.04	13.9	B
Herb Hill Road & Charles Street	EB	27	0.03	13.9	B
Herb Hill Road & Charles Street	WB	23	0.03	13.9	B
Bridge Street & School Street	NB	82	0.10	14.5	B
Bridge Street & School Street	SB	80	0.10	14.5	B
Bridge Street & School Street	EB	82	0.10	14.5	B
Bridge Street & School Street	WB	67	0.08	14.3	B
Pratt Boulevard & Pulaski Street	NB	308	0.38	17.8	B
Pratt Boulevard & Pulaski Street	SB	232	0.29	16.5	B
Pratt Boulevard & Pulaski Street	EB	261	0.32	17.0	B
Pratt Boulevard & Pulaski Street	WB	235	0.29	16.6	B
Glen Street & Pulaski Street	NB	89	0.11	14.6	B
Glen Street & Pulaski Street	SB	80	0.10	14.5	B
Glen Street & Pulaski Street	EB	86	0.11	14.6	B

Intersection	Approach	Vol (vph)	v/c	Delay (s)	LOS
Glen Street & Pulaski Street	WB	56	0.07	14.2	B
Brewster Street & School Street	NB	341	0.42	18.4	B
Brewster Street & School Street	SB	284	0.35	17.3	B
Brewster Street & School Street	EB	232	0.29	16.5	B
Brewster Street & School Street	WB	205	0.25	16.1	B
School Street & Highland Road	NB	29	0.04	13.9	B
School Street & Highland Road	SB	25	0.03	13.9	B
School Street & Highland Road	EB	24	0.03	13.9	B
School Street & Highland Road	WB	24	0.03	13.9	B
Shore Road & Glen Cove Avenue	NB	563	0.69	24.7	C
Shore Road & Glen Cove Avenue	SB	395	0.49	19.5	B
Shore Road & Glen Cove Avenue	EB	395	0.49	19.5	B
Shore Road & Glen Cove Avenue	WB	386	0.48	19.3	B
Glen Street & Knoll Place	NB	93	0.12	14.6	B
Glen Street & Knoll Place	SB	76	0.09	14.4	B
Glen Street & Knoll Place	EB	74	0.09	14.4	B
Glen Street & Knoll Place	WB	67	0.08	14.3	B
Pratt Boulevard & Town Path	NB	344	0.42	18.5	B
Pratt Boulevard & Town Path	SB	295	0.36	17.6	B
Pratt Boulevard & Town Path	EB	328	0.40	18.1	B
Pratt Boulevard & Town Path	WB	226	0.28	16.4	B
School Street & Brewster Street	NB	116	0.14	14.9	B
School Street & Brewster Street	SB	87	0.11	14.6	B
School Street & Brewster Street	EB	92	0.11	14.6	B
School Street & Brewster Street	WB	78	0.10	14.5	B
Landing Road & Roosevelt Street	NB	102	0.13	14.7	B
Landing Road & Roosevelt Street	SB	101	0.12	14.7	B
Landing Road & Roosevelt Street	EB	88	0.11	14.6	B
Landing Road & Roosevelt Street	WB	72	0.09	14.4	B
Glen Street & Town Path	NB	84	0.10	14.5	B
Glen Street & Town Path	SB	94	0.12	14.6	B
Glen Street & Town Path	EB	72	0.09	14.4	B
Glen Street & Town Path	WB	61	0.08	14.3	B
Burns Avenue & Glen Cove Avenue	NB	527	0.65	23.3	C
Burns Avenue & Glen Cove Avenue	SB	373	0.46	19.1	B
Burns Avenue & Glen Cove Avenue	EB	497	0.61	22.3	C
Burns Avenue & Glen Cove Avenue	WB	341	0.42	18.4	B
Dosoris Lane & Forest Avenue	NB	33	0.04	14.0	B
Dosoris Lane & Forest Avenue	SB	31	0.04	13.9	B

Intersection	Approach	Vol (vph)	v/c	Delay (s)	LOS
Dosoris Lane & Forest Avenue	EB	30	0.04	13.9	B
Dosoris Lane & Forest Avenue	WB	23	0.03	13.9	B
Glen Street & Hendrick Avenue	NB	52	0.06	14.2	B
Glen Street & Hendrick Avenue	SB	37	0.05	14.0	B
Glen Street & Hendrick Avenue	EB	47	0.06	14.1	B
Glen Street & Hendrick Avenue	WB	31	0.04	13.9	B
Forest Avenue & Elliott Place	NB	441	0.54	20.6	C
Forest Avenue & Elliott Place	SB	373	0.46	19.1	B
Forest Avenue & Elliott Place	EB	367	0.45	18.9	B
Forest Avenue & Elliott Place	WB	338	0.42	18.3	B
Cedar Swamp Road & Elm Avenue	NB	559	0.69	24.5	C
Cedar Swamp Road & Elm Avenue	SB	515	0.64	22.9	C
Cedar Swamp Road & Elm Avenue	EB	448	0.55	20.8	C
Cedar Swamp Road & Elm Avenue	WB	367	0.45	18.9	B

APPENDIX C — PROPOSED CONDITION CAPACITY ANALYSIS OUTPUT

Per-intersection screening-solver output (Webster/Akçelik control-delay model) for the Proposed Build condition (No-Build volumes plus project external trips at the assigned distribution).

Intersection	Approach	Vol (vph)	v/c	Delay (s)	LOS
Glen Cove Avenue & Charles Street	NB	588	0.73	25.9	C
Glen Cove Avenue & Charles Street	SB	449	0.55	20.9	C
Glen Cove Avenue & Charles Street	EB	488	0.60	22.0	C
Glen Cove Avenue & Charles Street	WB	370	0.46	19.0	B
Glen Cove Avenue & Brewster Street	NB	586	0.72	25.7	C
Glen Cove Avenue & Brewster Street	SB	467	0.58	21.4	C
Glen Cove Avenue & Brewster Street	EB	438	0.54	20.6	C
Glen Cove Avenue & Brewster Street	WB	350	0.43	18.6	B
Glen Cove Avenue & Continental Place	NB	90	0.11	14.6	B
Glen Cove Avenue & Continental Place	SB	79	0.10	14.5	B
Glen Cove Avenue & Continental Place	EB	105	0.13	14.8	B
Glen Cove Avenue & Continental Place	WB	79	0.10	14.5	B
Herb Hill Road & Brewster Street	NB	526	0.65	23.2	C
Herb Hill Road & Brewster Street	SB	498	0.62	22.3	C
Herb Hill Road & Brewster Street	EB	423	0.52	20.2	C
Herb Hill Road & Brewster Street	WB	354	0.44	18.7	B
Morris Avenue & Glen Cove Avenue	NB	566	0.70	24.8	C
Morris Avenue & Glen Cove Avenue	SB	479	0.59	21.7	C
Morris Avenue & Glen Cove Avenue	EB	419	0.52	20.1	C
Morris Avenue & Glen Cove Avenue	WB	334	0.41	18.3	B
Herb Hill Road & Charles Street	NB	49	0.06	14.1	B
Herb Hill Road & Charles Street	SB	32	0.04	13.9	B
Herb Hill Road & Charles Street	EB	51	0.06	14.2	B
Herb Hill Road & Charles Street	WB	62	0.08	14.3	B
Bridge Street & School Street	NB	120	0.15	15.0	B
Bridge Street & School Street	SB	106	0.13	14.8	B
Bridge Street & School Street	EB	82	0.10	14.5	B
Bridge Street & School Street	WB	77	0.09	14.5	B
Pratt Boulevard & Pulaski Street	NB	308	0.38	17.8	B
Pratt Boulevard & Pulaski Street	SB	232	0.29	16.5	B
Pratt Boulevard & Pulaski Street	EB	284	0.35	17.4	B
Pratt Boulevard & Pulaski Street	WB	235	0.29	16.6	B
Glen Street & Pulaski Street	NB	89	0.11	14.6	B
Glen Street & Pulaski Street	SB	94	0.12	14.7	B
Glen Street & Pulaski Street	EB	112	0.14	14.9	B

Intersection	Approach	Vol (vph)	v/c	Delay (s)	LOS
Glen Street & Pulaski Street	WB	67	0.08	14.3	B
Brewster Street & School Street	NB	360	0.44	18.8	B
Brewster Street & School Street	SB	298	0.37	17.6	B
Brewster Street & School Street	EB	232	0.29	16.5	B
Brewster Street & School Street	WB	207	0.26	16.1	B
School Street & Highland Road	NB	47	0.06	14.1	B
School Street & Highland Road	SB	38	0.05	14.0	B
School Street & Highland Road	EB	24	0.03	13.9	B
School Street & Highland Road	WB	28	0.03	13.9	B
Shore Road & Glen Cove Avenue	NB	576	0.71	25.3	C
Shore Road & Glen Cove Avenue	SB	427	0.53	20.3	C
Shore Road & Glen Cove Avenue	EB	410	0.51	19.9	B
Shore Road & Glen Cove Avenue	WB	386	0.48	19.3	B
Glen Street & Knoll Place	NB	93	0.12	14.6	B
Glen Street & Knoll Place	SB	81	0.10	14.5	B
Glen Street & Knoll Place	EB	85	0.10	14.5	B
Glen Street & Knoll Place	WB	73	0.09	14.4	B
Pratt Boulevard & Town Path	NB	344	0.42	18.5	B
Pratt Boulevard & Town Path	SB	295	0.36	17.6	B
Pratt Boulevard & Town Path	EB	350	0.43	18.6	B
Pratt Boulevard & Town Path	WB	226	0.28	16.4	B
School Street & Brewster Street	NB	148	0.18	15.3	B
School Street & Brewster Street	SB	117	0.14	14.9	B
School Street & Brewster Street	EB	92	0.11	14.6	B
School Street & Brewster Street	WB	78	0.10	14.5	B
Landing Road & Roosevelt Street	NB	111	0.14	14.9	B
Landing Road & Roosevelt Street	SB	109	0.13	14.8	B
Landing Road & Roosevelt Street	EB	88	0.11	14.6	B
Landing Road & Roosevelt Street	WB	73	0.09	14.4	B
Glen Street & Town Path	NB	94	0.12	14.7	B
Glen Street & Town Path	SB	115	0.14	14.9	B
Glen Street & Town Path	EB	72	0.09	14.4	B
Glen Street & Town Path	WB	61	0.08	14.3	B
Burns Avenue & Glen Cove Avenue	NB	540	0.67	23.8	C
Burns Avenue & Glen Cove Avenue	SB	388	0.48	19.4	B
Burns Avenue & Glen Cove Avenue	EB	497	0.61	22.3	C
Burns Avenue & Glen Cove Avenue	WB	341	0.42	18.4	B
Dosoris Lane & Forest Avenue	NB	33	0.04	14.0	B
Dosoris Lane & Forest Avenue	SB	31	0.04	13.9	B

Intersection	Approach	Vol (vph)	v/c	Delay (s)	LOS
Dosoris Lane & Forest Avenue	EB	63	0.08	14.3	B
Dosoris Lane & Forest Avenue	WB	53	0.07	14.2	B
Glen Street & Hendrick Avenue	NB	52	0.06	14.2	B
Glen Street & Hendrick Avenue	SB	50	0.06	14.1	B
Glen Street & Hendrick Avenue	EB	47	0.06	14.1	B
Glen Street & Hendrick Avenue	WB	31	0.04	13.9	B
Forest Avenue & Elliott Place	NB	441	0.54	20.6	C
Forest Avenue & Elliott Place	SB	373	0.46	19.1	B
Forest Avenue & Elliott Place	EB	399	0.49	19.6	B
Forest Avenue & Elliott Place	WB	367	0.45	18.9	B
Cedar Swamp Road & Elm Avenue	NB	559	0.69	24.5	C
Cedar Swamp Road & Elm Avenue	SB	528	0.65	23.3	C
Cedar Swamp Road & Elm Avenue	EB	448	0.55	20.8	C
Cedar Swamp Road & Elm Avenue	WB	367	0.45	18.9	B

APPENDIX D — CRASH ANALYSIS DIAGRAMS AND TABLES

Crash analysis diagrams and tables are not produced by this screening analysis. When the §4.0 crash-analysis section is expanded for formal submittal, the following NYSDOT forms are required: TE-156a (Collision Diagram), TE-164a (Safety Benefits Evaluation Form), TE-204a (Accident Rate Summary), and TE-213 (HAL / PIL / SDL Screening Output). SIMS output and statewide-average rate references should be obtained from the controlling Regional Traffic Office per HDM Chapter 5 §5.3.4 and the NYSDOT Office of Modal Safety statewide rate tables (https://www.dot.ny.gov/divisions/operating/osss/highway/accident-rates).

This report is based on the NYSDOT TIS Shell revised on 9/16/2014.

FOUR-STEP TRAVEL DEMAND MODEL

Off-site project trips are distributed and assigned with the four-step urban transportation modeling process (FHWA; NCHRP Report 716, Travel Demand Forecasting). Each step below shows what the engine computed for this site.

Step 1 — Trip Generation

Public-data screening trip rates (SANDAG 2002, corroborated by NHTS 2017 / NCHRP 716) applied to the proposed land use give the site's productions.

Land use	LU 820 — Shopping Center / Retail
Size	85 1,000 sqft GLA
Daily trips (productions)	6,800
PM peak-hour trips	680 (326 in / 354 out)

Step 2 — Trip Distribution (Gravity Model)

A production-constrained gravity model allocates trips to surrounding signals: $T_j = P \cdot (A_j \cdot F_j) / \Sigma (A_x \cdot F_x)$. Attractiveness A_j is each signal's through-volume; the friction factor F_j is a gamma function $F = a \cdot t^b \cdot e^{(c \cdot t)}$ — the form recommended by NCHRP Report 716 — on the travel time t to the signal, with home-based-work shape parameters b=-0.02, c=-0.123 (consistent with NCHRP 716's sampled MPO gravity parameters; the scale constant a cancels in the share normalization).

Signal (attraction zone)	Dist mi	Time min	PM trips	Share
Glen Cove Avenue & Charles Street	0.01	0.1	156	14.0%
Glen Cove Avenue & Brewster Street	0.04	0.1	106	9.5%
Herb Hill Road & Charles Street	0.16	0.4	75	6.7%
Bridge Street & School Street	0.16	0.4	75	6.7%
Herb Hill Road & Brewster Street	0.10	0.2	62	5.6%
School Street & Brewster Street	0.43	1.0	62	5.6%
Dosoris Lane & Forest Avenue	0.54	1.3	62	5.6%
Forest Avenue & Elliott Place	0.67	1.6	62	5.6%
Morris Avenue & Glen Cove Avenue	0.14	0.3	60	5.4%
Shore Road & Glen Cove Avenue	0.37	0.9	60	5.4%
Glen Street & Pulaski Street	0.24	0.6	51	4.6%
Glen Cove Avenue & Continental Place	0.10	0.2	44	3.9%

Top 12 distribution zones by assigned PM-peak trips shown; the full set is in the capacity appendix.

Step 3 — Mode Choice

Auto-mode share 35% applied via a binary logit $P(auto) = 1 / (1 + e^{-(ASC - \lambda \cdot \Delta GC)})$ calibrated to the metro's measured auto-mode share (ACS B08301) and shifted by site urbanity (local density), so a denser, more transit-served site splits further from auto than a greenfield parcel in the same metro. The remaining 65% of trips (transit, walking, cycling) do not load the off-site roadway network; only auto trips are carried into Step 4.

Step 4 — Route Assignment

Project trips are routed over the local road network by shortest path and loaded with a capacity-constrained BPR equilibrium (volume-delay $t = t_0 \cdot [1 + 0.15 \cdot (v/c)^4]$, Method of Successive Averages), so congested links shift trips toward alternative routes. 100% of study signals lie on a modeled route; highest loaded-link v/c: 0.17. Corridors carrying the project trips:

Corridor (road class)	Loaded length mi	PM project vph	v/c
Minor arterial	1.98	106	0.17
Principal arterial	0.44	64	0.16
Collector	0.80	5	0.15

Network shortest-path assignment over the OSM road graph within the study area; per-signal v/c, delay, LOS, and queue are in the capacity appendix.

Conserved assignment: project trips routed to 15 cordon gateways on the study boundary (weighted by the directional distribution above). Flow conservation verified at render input: Σ entering = Σ leaving at 382 network junctions, max imbalance 0.00 veh. 14 study intersection(s) carry path-derived movements; 8 retain the geometric octant model (labeled per intersection).

APPENDIX — INTERSECTION CAPACITY ANALYSIS WORKSHEETS

One worksheet per study intersection: a turning-movement diagram for each analyzed peak period plus the per-approach capacity table. Analysis uses the openly-published Webster/Akcelik signalized control-delay method; v/c, control delay (sec/veh), Level of Service, and 95th-percentile back-of-queue (ft) are reported for the Existing (No-Build) and Build conditions.

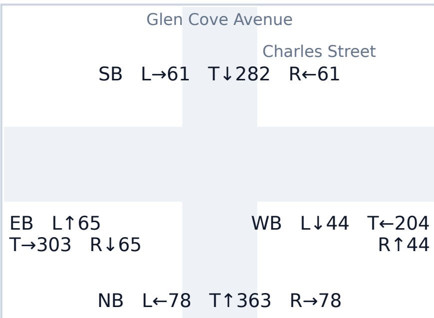
Background turning-movement volumes in the diagrams are distributed from each approach total using an estimated 15/70/15 (Left/Through/Right) split. Project-trip movements are assigned geometrically from the study's directional trip distribution (see each worksheet's Affected movements table). Replace both with measured turning-movement counts (TMCs) before a formal submittal.

A.1 Glen Cove Avenue & Charles Street

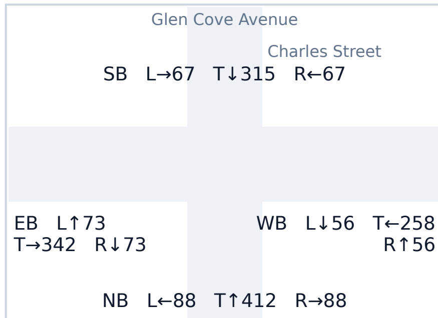
Signal ID	new-york-14074139992
Location	NE New York-Newark-Jersey City · 0.01 mi from site
Added PM peak trips	156
Existing / No-Build (PM)	LOS D · 48.4 s/veh · v/c 0.97
Build (PM)	LOS E · 71.0 s/veh · v/c 1.05
Δ control delay	+22.6 s/veh (LOS change)
Mitigation	Moderate: extend critical-phase green time and consider a protected-only left-turn phase to absorb the new demand without queue spillback.

Turning-Movement Diagrams (Build condition)

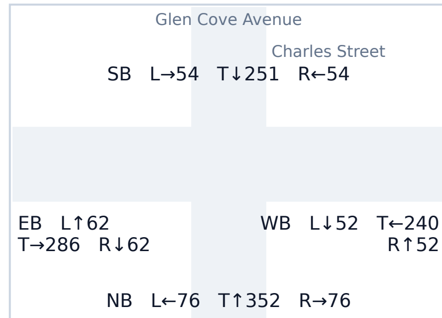
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	541	47	588	0.67 → 0.73	23.8 → 25.9	C → C	496
SB	449	0	449	0.55 → 0.55	20.9 → 20.9	C → C	339
EB	460	28	488	0.57 → 0.60	21.2 → 22.0	C → C	380
WB	289	81	370	0.36 → 0.46	17.4 → 19.0	B → B	264

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Left	51	33%
NB Right	47	30%
WB Through	30	19%
EB Through	28	18%

Source: geometric octant model (this signal did not resolve to a junction on the modeled road network — no junction within 100 m, or its minor legs are below the network's road-class floor).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split,

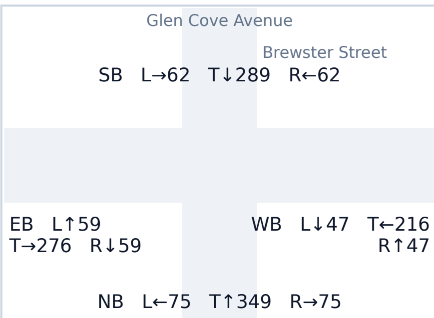
not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.2 Glen Cove Avenue & Brewster Street

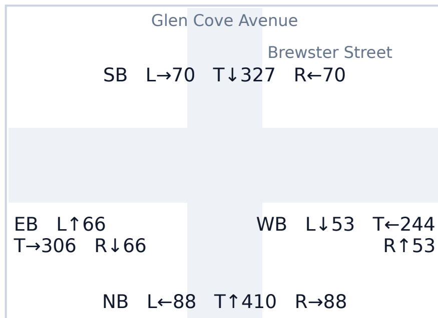
Signal ID	new-york-16596
Location	NE New York-Newark-Jersey City · 0.04 mi from site
Added PM peak trips	106
Existing / No-Build (PM)	LOS D · 48.0 s/veh · v/c 0.96
Build (PM)	LOS E · 62.0 s/veh · v/c 1.02
Δ control delay	+14.0 s/veh (LOS change)
Mitigation	Moderate: extend critical-phase green time and consider a protected-only left-turn phase to absorb the new demand without queue spillback.

Turning-Movement Diagrams (Build condition)

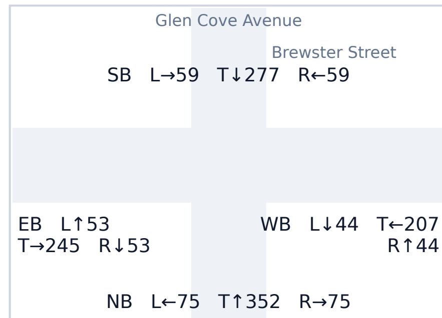
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	531	55	586	0.66 → 0.72	23.4 → 25.7	C → C	493
SB	437	30	467	0.54 → 0.58	20.5 → 21.4	C → C	357
EB	438	0	438	0.54 → 0.54	20.6 → 20.6	C → C	328
WB	329	21	350	0.41 → 0.43	18.2 → 18.6	B → B	247

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
NB Through	32	30%
SB Through	30	28%
NB Right	23	22%
WB Left	21	20%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split,

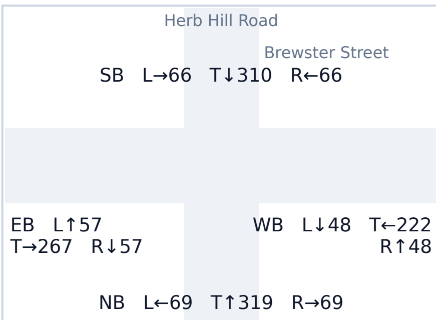
not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.3 Herb Hill Road & Brewster Street

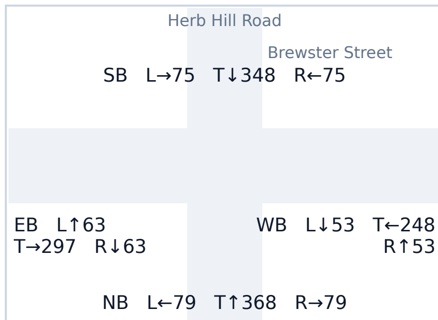
Signal ID	new-york-12426
Location	NE New York-Newark-Jersey City · 0.10 mi from site
Added PM peak trips	62
Existing / No-Build (PM)	LOS D · 48.4 s/veh · v/c 0.97
Build (PM)	LOS E · 56.3 s/veh · v/c 1.00
Δ control delay	+7.9 s/veh (LOS change)
Mitigation	Moderate: extend critical-phase green time and consider a protected-only left-turn phase to absorb the new demand without queue spillback.

Turning-Movement Diagrams (Build condition)

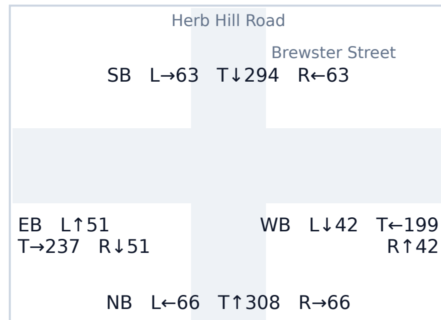
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	493	32	526	0.61 → 0.65	22.1 → 23.2	C → C	421
SB	469	30	498	0.58 → 0.62	21.4 → 22.3	C → C	391
EB	423	0	423	0.52 → 0.52	20.2 → 20.2	C → C	314
WB	354	0	354	0.44 → 0.44	18.7 → 18.7	B → B	250

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
NB Through	32	52%
SB Through	30	48%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

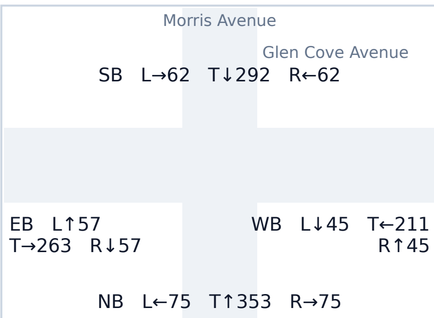
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.4 Morris Avenue & Glen Cove Avenue

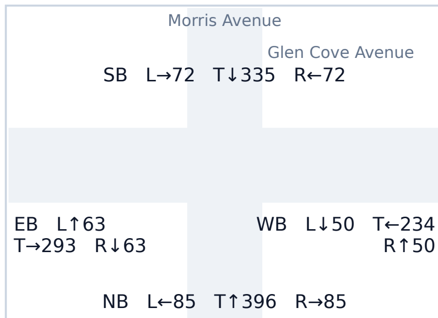
Signal ID	new-york-13514
Location	NE New York-Newark-Jersey City · 0.14 mi from site
Added PM peak trips	60
Existing / No-Build (PM)	LOS D · 48.4 s/veh · v/c 0.97
Build (PM)	LOS E · 56.1 s/veh · v/c 1.00
Δ control delay	+7.7 s/veh (LOS change)
Mitigation	Moderate: extend critical-phase green time and consider a protected-only left-turn phase to absorb the new demand without queue spillback.

Turning-Movement Diagrams (Build condition)

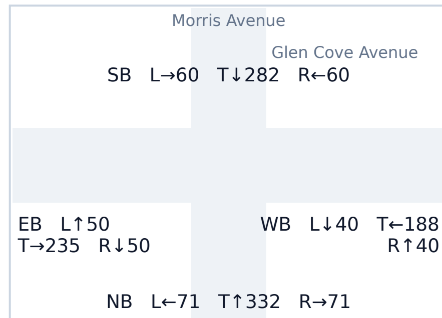
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	537	29	566	0.66 → 0.70	23.7 → 24.8	C → C	469
SB	448	31	479	0.55 → 0.59	20.8 → 21.7	C → C	371
EB	419	0	419	0.52 → 0.52	20.1 → 20.1	C → C	310
WB	334	0	334	0.41 → 0.41	18.3 → 18.3	B → B	233

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
SB Through	31	52%
NB Through	29	48%

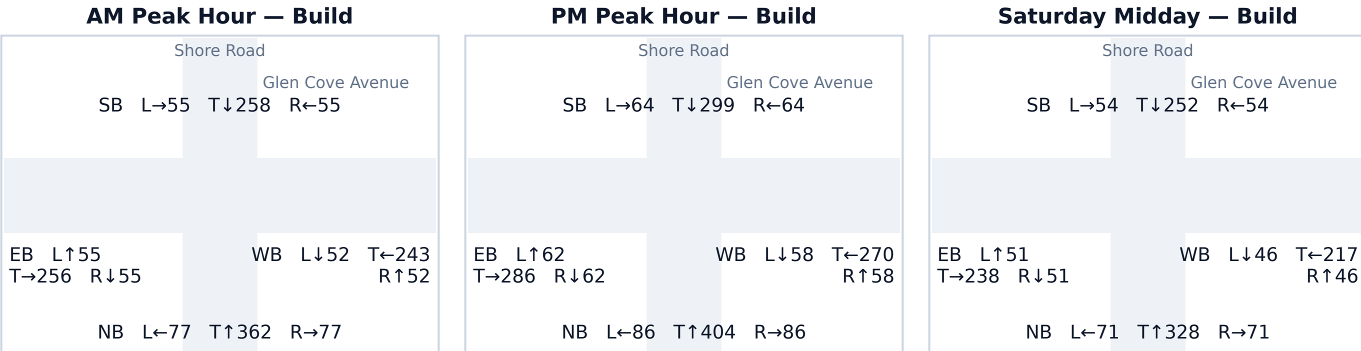
Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.5 Shore Road & Glen Cove Avenue

Signal ID	new-york-13099
Location	NE New York-Newark-Jersey City · 0.37 mi from site
Added PM peak trips	60
Existing / No-Build (PM)	LOS D · 48.4 s/veh · v/c 0.97
Build (PM)	LOS E · 56.1 s/veh · v/c 1.00
Δ control delay	+7.7 s/veh (LOS change)
Mitigation	Moderate: extend critical-phase green time and consider a protected-only left-turn phase to absorb the new demand without queue spillback.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	563	13	576	0.69 → 0.71	24.7 → 25.3	C → C	481
SB	395	31	427	0.49 → 0.53	19.5 → 20.3	B → C	317
EB	395	16	410	0.49 → 0.51	19.5 → 19.9	B → B	301
WB	386	0	386	0.48 → 0.48	19.3 → 19.3	B → B	279

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
SB Right	17	28%
EB Left	16	27%
SB Through	14	23%
NB Through	13	22%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split,

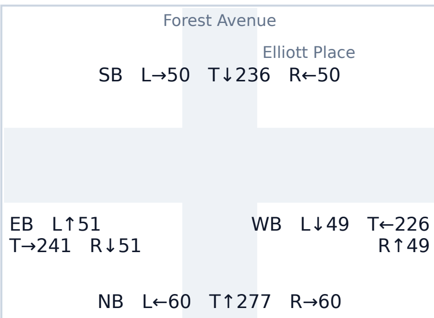
not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.6 Forest Avenue & Elliott Place

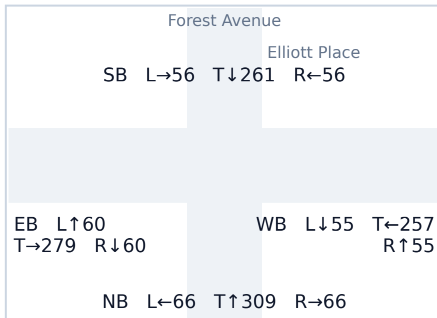
Signal ID	new-york-14799
Location	NE New York-Newark-Jersey City · 0.67 mi from site
Added PM peak trips	62
Existing / No-Build (PM)	LOS C · 32.4 s/veh · v/c 0.84
Build (PM)	LOS D · 35.5 s/veh · v/c 0.88
Δ control delay	+3.1 s/veh (LOS change)
Mitigation	Minor: signal-timing optimization (shift 3–5s of green to the critical phase) is sufficient. No geometric change required.

Turning-Movement Diagrams (Build condition)

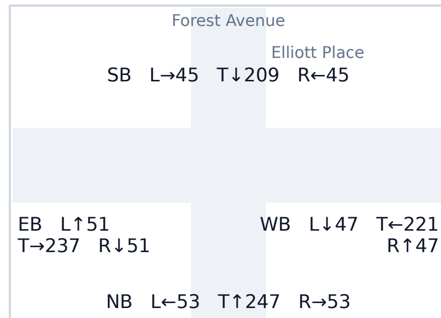
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	441	0	441	0.54 → 0.54	20.6 → 20.6	C → C	331
SB	373	0	373	0.46 → 0.46	19.1 → 19.1	B → B	267
EB	367	32	399	0.45 → 0.49	18.9 → 19.6	B → B	291
WB	338	30	367	0.42 → 0.45	18.3 → 18.9	B → B	262

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
EB Through	32	52%
WB Through	30	48%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

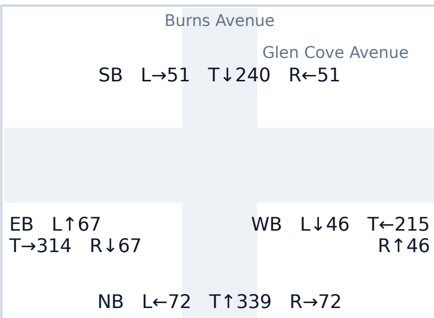
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.7 Burns Avenue & Glen Cove Avenue

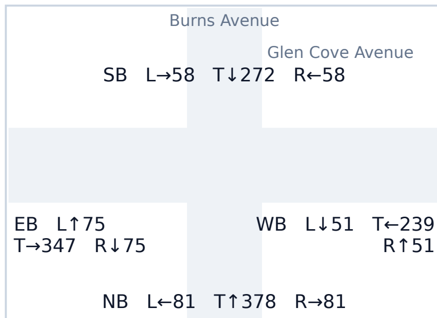
Signal ID	new-york-13589
Location	NE New York-Newark-Jersey City · 0.50 mi from site
Added PM peak trips	28
Existing / No-Build (PM)	LOS D · 48.4 s/veh · v/c 0.97
Build (PM)	LOS D · 51.8 s/veh · v/c 0.98
Δ control delay	+3.4 s/veh
Mitigation	Minor: signal-timing optimization (shift 3–5s of green to the critical phase) is sufficient. No geometric change required.

Turning-Movement Diagrams (Build condition)

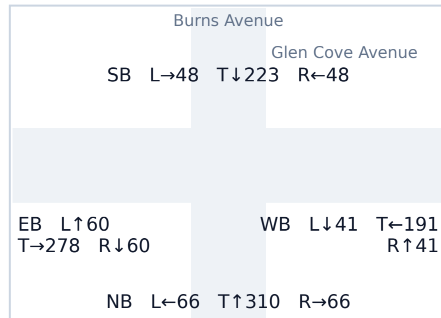
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	527	13	540	0.65 → 0.67	23.3 → 23.8	C → C	438
SB	373	15	388	0.46 → 0.48	19.1 → 19.4	B → B	280
EB	497	0	497	0.61 → 0.61	22.3 → 22.3	C → C	390
WB	341	0	341	0.42 → 0.42	18.4 → 18.4	B → B	239

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
SB Through	15	54%
NB Through	13	46%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

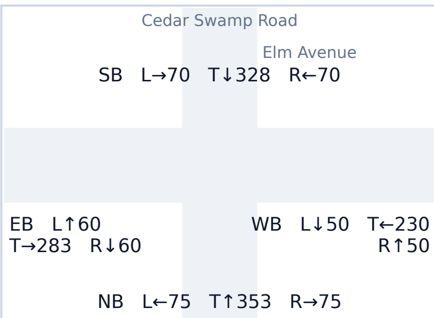
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.8 Cedar Swamp Road & Elm Avenue

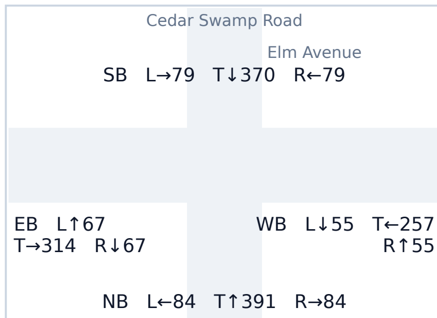
Signal ID	new-york-13215
Location	NE New York-Newark-Jersey City · 0.72 mi from site
Added PM peak trips	13
Existing / No-Build (PM)	LOS E · 69.9 s/veh · v/c 1.05
Build (PM)	LOS E · 72.2 s/veh · v/c 1.06
Δ control delay	+2.3 s/veh
Mitigation	Moderate: extend critical-phase green time and consider a protected-only left-turn phase to absorb the new demand without queue spillback.

Turning-Movement Diagrams (Build condition)

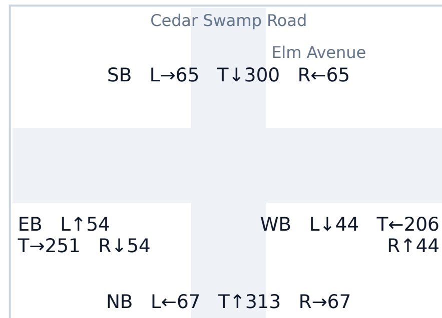
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	559	0	559	0.69 → 0.69	24.5 → 24.5	C → C	460
SB	515	13	528	0.64 → 0.65	22.9 → 23.3	C → C	423
EB	448	0	448	0.55 → 0.55	20.8 → 20.8	C → C	338
WB	367	0	367	0.45 → 0.45	18.9 → 18.9	B → B	262

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
SB Through	13	100%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

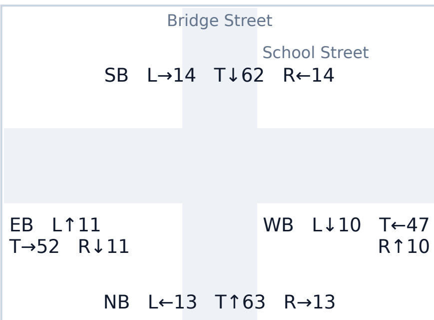
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.9 Bridge Street & School Street

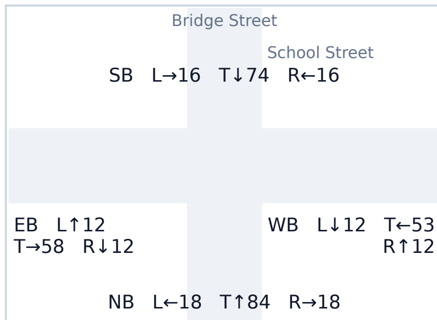
Signal ID	new-york-13429
Location	NE New York-Newark-Jersey City · 0.16 mi from site
Added PM peak trips	75
Existing / No-Build (PM)	LOS B · 15.2 s/veh · v/c 0.17
Build (PM)	LOS B · 15.7 s/veh · v/c 0.21
Δ control delay	+0.5 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)

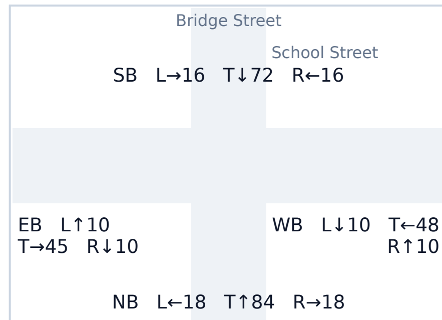
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	82	39	120	0.10 → 0.15	14.5 → 15.0	B → B	73
SB	80	26	106	0.10 → 0.13	14.5 → 14.8	B → B	64
EB	82	0	82	0.10 → 0.10	14.5 → 14.5	B → B	49
WB	67	10	77	0.08 → 0.09	14.3 → 14.5	B → B	46

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
NB Through	28	37%
SB Through	26	35%
NB Right	11	15%
WB Left	10	13%

Source: geometric octant model (this signal did not resolve to a junction on the modeled road network — no junction within 100 m, or its minor legs are below the network's road-class floor).

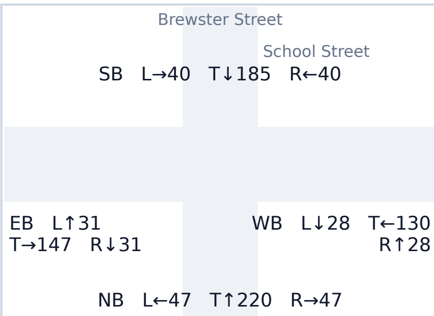
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.10 Brewster Street & School Street

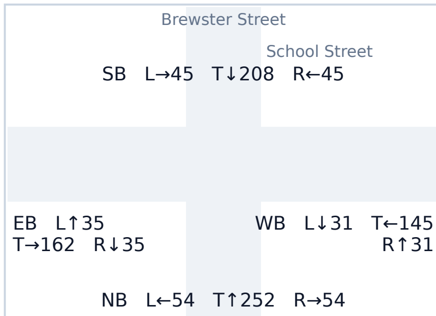
Signal ID	new-york-18023
Location	NE New York-Newark-Jersey City · 0.30 mi from site
Added PM peak trips	36
Existing / No-Build (PM)	LOS C · 21.7 s/veh · v/c 0.59
Build (PM)	LOS C · 22.2 s/veh · v/c 0.61
Δ control delay	+0.5 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)

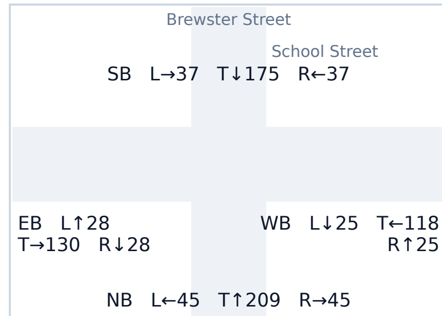
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	341	19	360	0.42 → 0.44	18.4 → 18.8	B → B	255
SB	284	14	298	0.35 → 0.37	17.3 → 17.6	B → B	203
EB	232	0	232	0.29 → 0.29	16.5 → 16.5	B → B	151
WB	205	3	207	0.25 → 0.26	16.1 → 16.1	B → B	133

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
NB Through	16	44%
SB Through	14	39%
NB Right	3	8%
WB Left	3	8%

Source: geometric octant model (this signal did not resolve to a junction on the modeled road network — no junction within 100 m, or its minor legs are below the network's road-class floor).

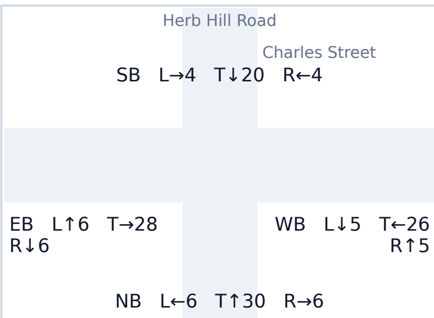
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.11 Herb Hill Road & Charles Street

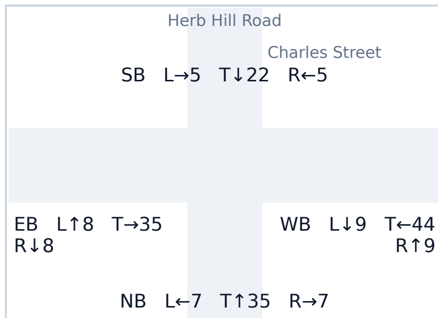
Signal ID	new-york-12874
Location	NE New York-Newark-Jersey City · 0.16 mi from site
Added PM peak trips	75
Existing / No-Build (PM)	LOS B · 14.2 s/veh · v/c 0.07
Build (PM)	LOS B · 14.6 s/veh · v/c 0.11
Δ control delay	+0.4 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)

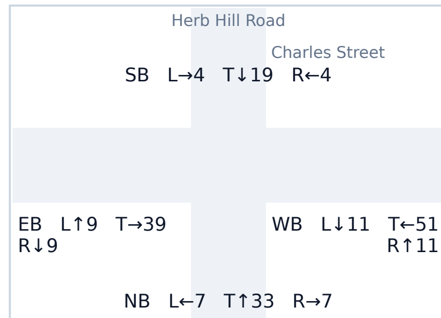
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	39	10	49	0.05 → 0.06	14.0 → 14.1	B → B	29
SB	29	3	32	0.04 → 0.04	13.9 → 13.9	B → B	18
EB	27	23	51	0.03 → 0.06	13.9 → 14.2	B → B	29
WB	23	39	62	0.03 → 0.08	13.9 → 14.3	B → B	37

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Through	25	33%
EB Through	23	31%
WB Left	11	15%
NB Right	10	13%
WB Right	3	4%
SB Left	3	4%

Source: geometric octant model (this signal did not resolve to a junction on the modeled road network — no junction within 100 m, or its minor legs are below the network's road-class floor).

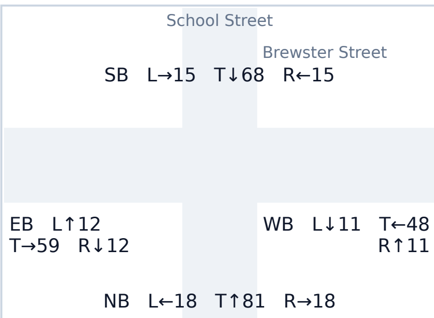
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.12 School Street & Brewster Street

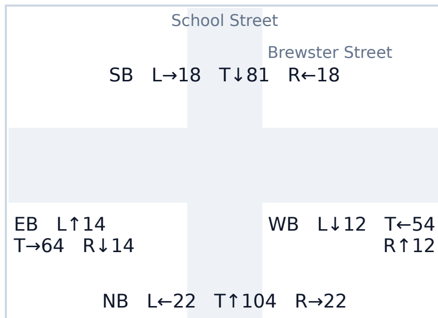
Signal ID	new-york-13699
Location	NE New York-Newark-Jersey City · 0.43 mi from site
Added PM peak trips	62
Existing / No-Build (PM)	LOS B · 15.6 s/veh · v/c 0.21
Build (PM)	LOS B · 16.0 s/veh · v/c 0.24
Δ control delay	+0.4 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)

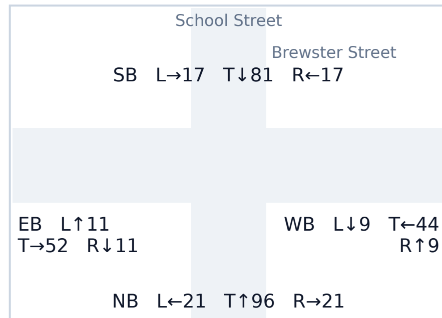
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	116	32	148	0.14 → 0.18	14.9 → 15.3	B → B	91
SB	87	30	117	0.11 → 0.14	14.6 → 14.9	B → B	71
EB	92	0	92	0.11 → 0.11	14.6 → 14.6	B → B	55
WB	78	0	78	0.10 → 0.10	14.5 → 14.5	B → B	46

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
NB Through	32	52%
SB Through	30	48%

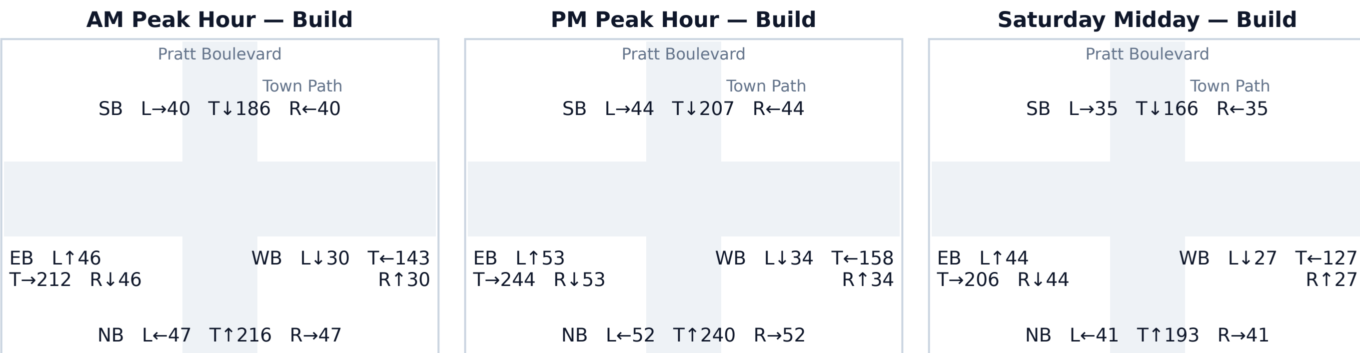
Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.13 Pratt Boulevard & Town Path

Signal ID	new-york-13270
Location	NE New York-Newark-Jersey City · 0.42 mi from site
Added PM peak trips	23
Existing / No-Build (PM)	LOS C · 23.6 s/veh · v/c 0.66
Build (PM)	LOS C · 24.0 s/veh · v/c 0.68
Δ control delay	+0.4 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	344	0	344	0.42 → 0.42	18.5 → 18.5	B → B	241
SB	295	0	295	0.36 → 0.36	17.6 → 17.6	B → B	200
EB	328	23	350	0.40 → 0.43	18.1 → 18.6	B → B	247
WB	226	0	226	0.28 → 0.28	16.4 → 16.4	B → B	146

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
EB Through	23	100%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

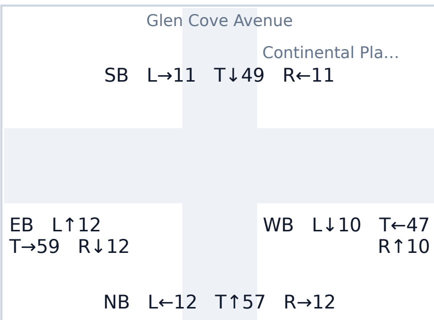
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.14 Glen Cove Avenue & Continental Place

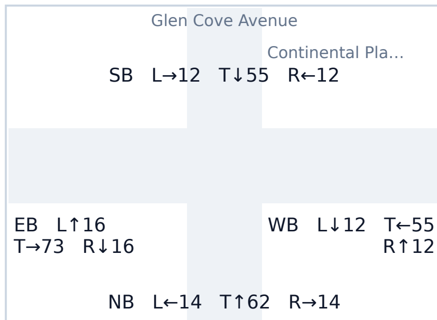
Signal ID	new-york-12469
Location	NE New York-Newark-Jersey City · 0.10 mi from site
Added PM peak trips	44
Existing / No-Build (PM)	LOS B · 15.2 s/veh · v/c 0.17
Build (PM)	LOS B · 15.5 s/veh · v/c 0.20
Δ control delay	+0.3 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)

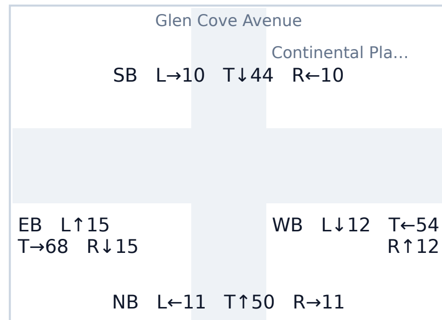
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	90	0	90	0.11 → 0.11	14.6 → 14.6	B → B	54
SB	79	0	79	0.10 → 0.10	14.5 → 14.5	B → B	47
EB	82	23	105	0.10 → 0.13	14.5 → 14.8	B → B	63
WB	58	21	79	0.07 → 0.10	14.2 → 14.5	B → B	47

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
EB Through	23	52%
WB Through	21	48%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

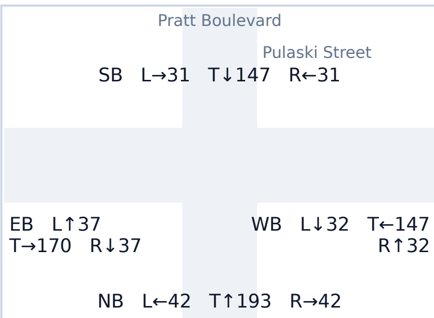
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.15 Pratt Boulevard & Pulaski Street

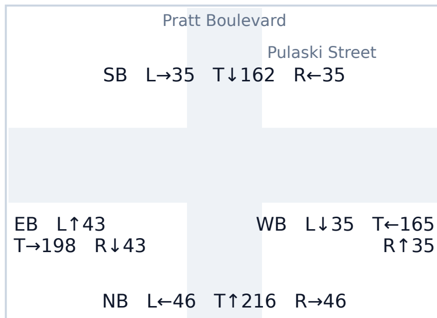
Signal ID	new-york-18387
Location	NE New York-Newark-Jersey City · 0.22 mi from site
Added PM peak trips	23
Existing / No-Build (PM)	LOS C · 21.3 s/veh · v/c 0.58
Build (PM)	LOS C · 21.6 s/veh · v/c 0.59
Δ control delay	+0.3 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)

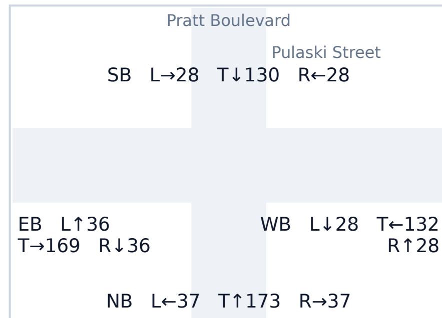
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	308	0	308	0.38 → 0.38	17.8 → 17.8	B → B	211
SB	232	0	232	0.29 → 0.29	16.5 → 16.5	B → B	151
EB	261	23	284	0.32 → 0.35	17.0 → 17.4	B → B	191
WB	235	0	235	0.29 → 0.29	16.6 → 16.6	B → B	153

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
EB Through	23	100%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

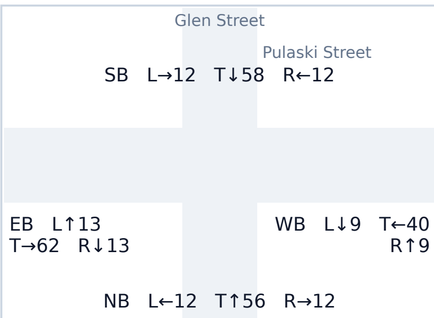
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.16 Glen Street & Pulaski Street

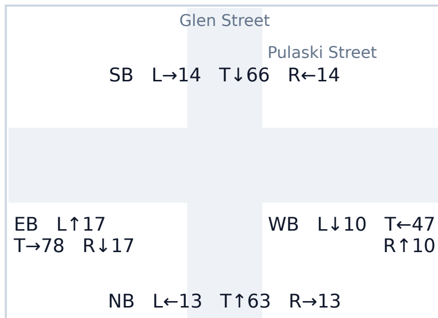
Signal ID	new-york-13273
Location	NE New York-Newark-Jersey City · 0.24 mi from site
Added PM peak trips	51
Existing / No-Build (PM)	LOS B · 15.2 s/veh · v/c 0.17
Build (PM)	LOS B · 15.5 s/veh · v/c 0.20
Δ control delay	+0.3 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)

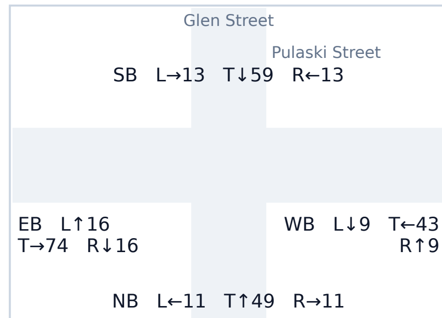
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	89	0	89	0.11 → 0.11	14.6 → 14.6	B → B	53
SB	80	14	94	0.10 → 0.12	14.5 → 14.7	B → B	56
EB	86	26	112	0.11 → 0.14	14.6 → 14.9	B → B	68
WB	56	11	67	0.07 → 0.08	14.2 → 14.3	B → B	39

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
EB Left	15	29%
SB Right	14	27%
EB Through	11	22%
WB Through	11	22%

Source: geometric octant model (this signal did not resolve to a junction on the modeled road network — no junction within 100 m, or its minor legs are below the network's road-class floor).

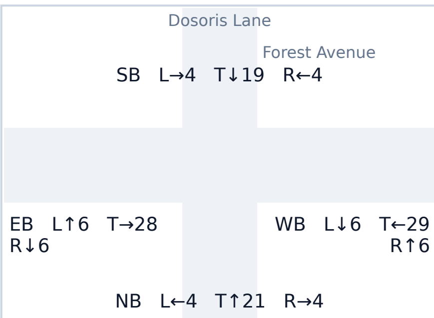
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.17 Dosoris Lane & Forest Avenue

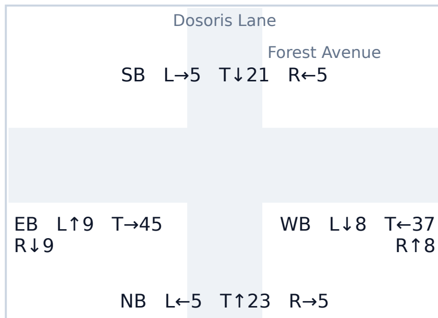
Signal ID	new-york-14798
Location	NE New York-Newark-Jersey City · 0.54 mi from site
Added PM peak trips	62
Existing / No-Build (PM)	LOS B · 14.2 s/veh · v/c 0.06
Build (PM)	LOS B · 14.5 s/veh · v/c 0.10
Δ control delay	+0.3 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)

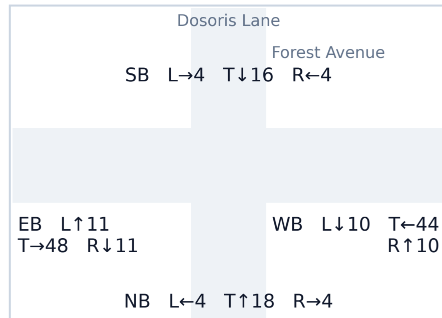
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	33	0	33	0.04 → 0.04	14.0 → 14.0	B → B	19
SB	31	0	31	0.04 → 0.04	13.9 → 13.9	B → B	18
EB	30	32	63	0.04 → 0.08	13.9 → 14.3	B → B	37
WB	23	30	53	0.03 → 0.07	13.9 → 14.2	B → B	31

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
EB Through	32	52%
WB Through	30	48%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

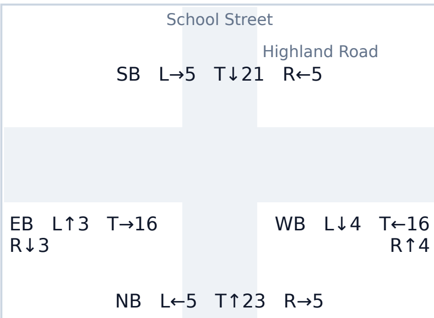
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.18 School Street & Highland Road

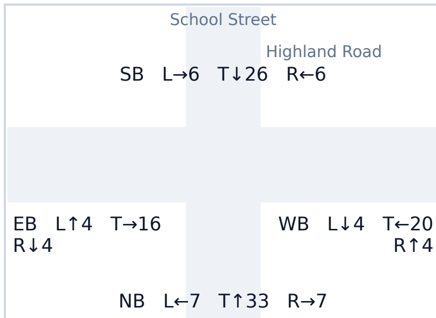
Signal ID	new-york-12696
Location	NE New York-Newark-Jersey City · 0.31 mi from site
Added PM peak trips	35
Existing / No-Build (PM)	LOS B · 14.1 s/veh · v/c 0.06
Build (PM)	LOS B · 14.3 s/veh · v/c 0.08
Δ control delay	+0.2 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)

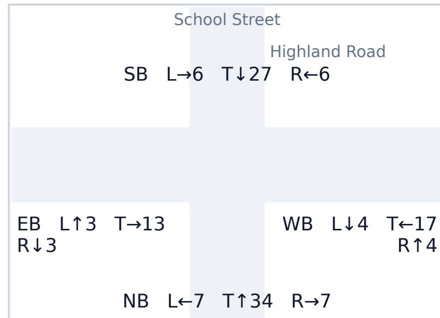
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	29	18	47	0.04 → 0.06	13.9 → 14.1	B → B	27
SB	25	13	38	0.03 → 0.05	13.9 → 14.0	B → B	22
EB	24	0	24	0.03 → 0.03	13.9 → 13.9	B → B	14
WB	24	4	28	0.03 → 0.03	13.9 → 13.9	B → B	16

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
NB Through	14	40%
SB Through	13	37%
NB Right	4	11%
WB Left	4	11%

Source: geometric octant model (this signal did not resolve to a junction on the modeled road network — no junction within 100 m, or its minor legs are below the network's road-class floor).

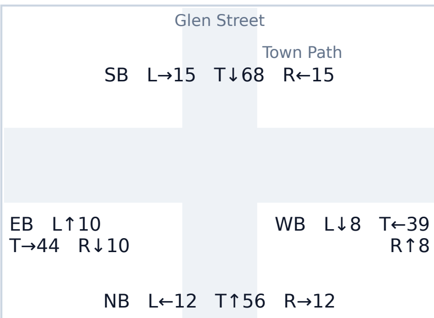
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.19 Glen Street & Town Path

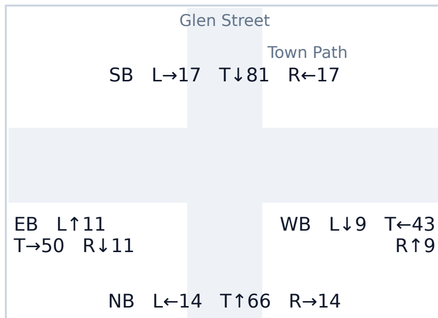
Signal ID	new-york-13271
Location	NE New York-Newark-Jersey City · 0.50 mi from site
Added PM peak trips	31
Existing / No-Build (PM)	LOS B · 15.2 s/veh · v/c 0.17
Build (PM)	LOS B · 15.4 s/veh · v/c 0.19
Δ control delay	+0.2 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)

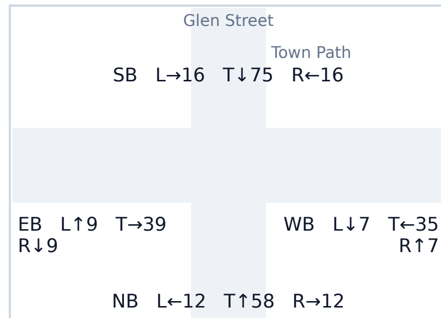
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	84	10	94	0.10 → 0.12	14.5 → 14.7	B → B	57
SB	94	21	115	0.12 → 0.14	14.6 → 14.9	B → B	70
EB	72	0	72	0.09 → 0.09	14.4 → 14.4	B → B	42
WB	61	0	61	0.08 → 0.08	14.3 → 14.3	B → B	36

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
SB Through	21	68%
NB Through	10	32%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

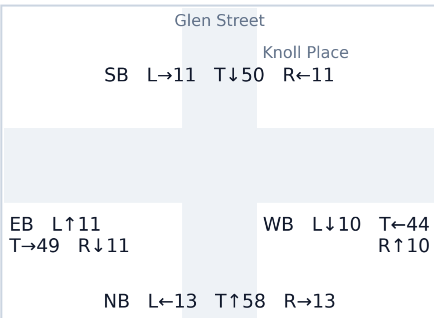
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.20 Glen Street & Knoll Place

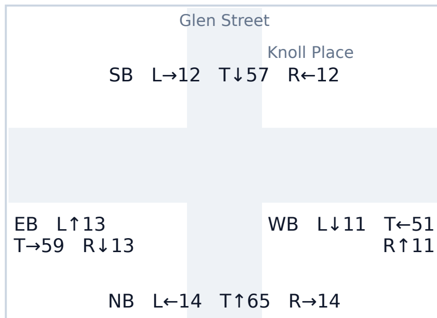
Signal ID	new-york-18321
Location	NE New York-Newark-Jersey City · 0.41 mi from site
Added PM peak trips	21
Existing / No-Build (PM)	LOS B · 15.2 s/veh · v/c 0.17
Build (PM)	LOS B · 15.3 s/veh · v/c 0.18
Δ control delay	+0.1 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)

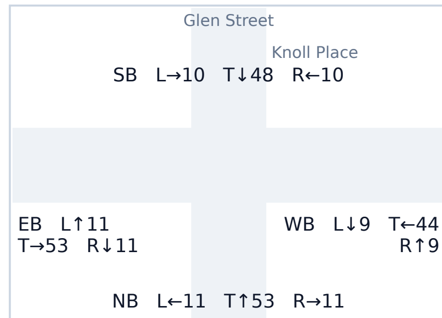
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	93	0	93	0.12 → 0.12	14.6 → 14.6	B → B	56
SB	76	4	81	0.09 → 0.10	14.4 → 14.5	B → B	48
EB	74	11	85	0.09 → 0.10	14.4 → 14.5	B → B	51
WB	67	6	73	0.08 → 0.09	14.3 → 14.4	B → B	43

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
EB Through	6	29%
WB Through	6	29%
EB Left	5	24%
SB Right	4	19%

Source: geometric octant model (this signal did not resolve to a junction on the modeled road network — no junction within 100 m, or its minor legs are below the network's road-class floor).

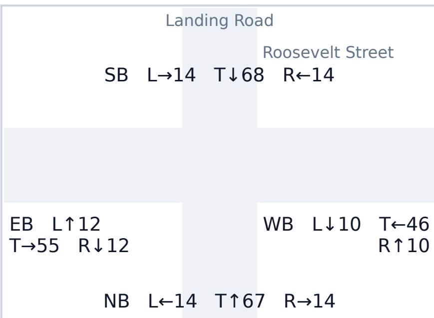
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.21 Landing Road & Roosevelt Street

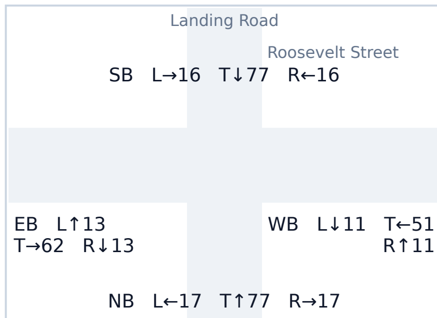
Signal ID	new-york-16650
Location	NE New York-Newark-Jersey City · 0.44 mi from site
Added PM peak trips	18
Existing / No-Build (PM)	LOS B · 15.5 s/veh · v/c 0.20
Build (PM)	LOS B · 15.6 s/veh · v/c 0.21
Δ control delay	+0.1 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)

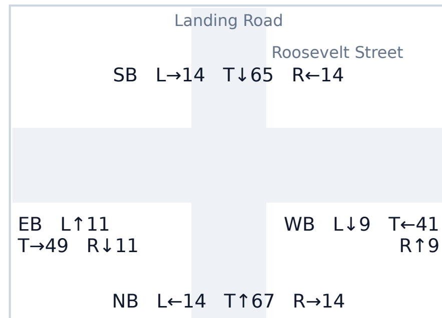
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	102	9	111	0.13 → 0.14	14.7 → 14.9	B → B	67
SB	101	8	109	0.12 → 0.13	14.7 → 14.8	B → B	66
EB	88	0	88	0.11 → 0.11	14.6 → 14.6	B → B	53
WB	72	1	73	0.09 → 0.09	14.4 → 14.4	B → B	43

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
NB Through	8	44%
SB Through	8	44%
NB Right	1	6%
WB Left	1	6%

Source: geometric octant model (this signal did not resolve to a junction on the modeled road network — no junction within 100 m, or its minor legs are below the network's road-class floor).

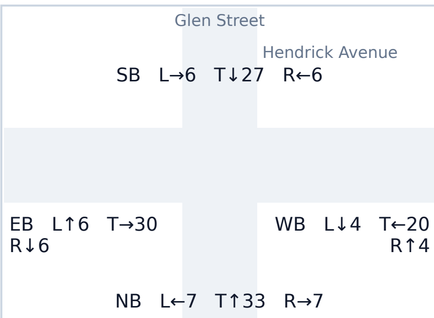
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.22 Glen Street & Hendrick Avenue

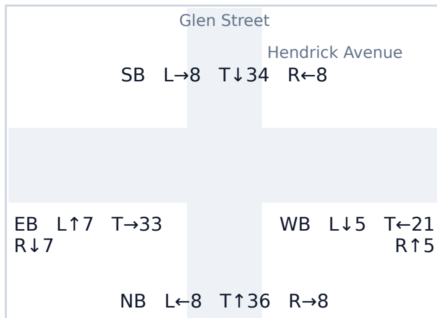
Signal ID	new-york-13501
Location	NE New York-Newark-Jersey City · 0.58 mi from site
Added PM peak trips	13
Existing / No-Build (PM)	LOS B · 14.4 s/veh · v/c 0.09
Build (PM)	LOS B · 14.5 s/veh · v/c 0.10
Δ control delay	+0.1 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)

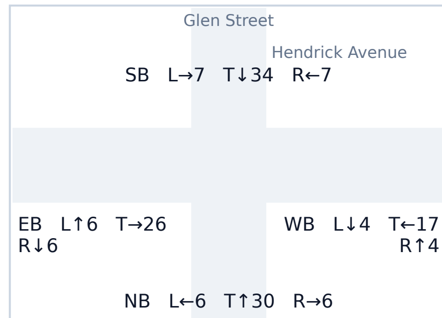
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	52	0	52	0.06 → 0.06	14.2 → 14.2	B → B	31
SB	37	13	50	0.05 → 0.06	14.0 → 14.1	B → B	29
EB	47	0	47	0.06 → 0.06	14.1 → 14.1	B → B	27
WB	31	0	31	0.04 → 0.04	13.9 → 13.9	B → B	18

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
SB Through	13	100%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.